

Review

Algal-Based Carbonaceous Materials for Environmental Remediation: Advances in Wastewater Treatment, Carbon Sequestration, and Biofuel Applications

Lázaro Adrián González Fernández ^{1,2,*} , Nahum Andrés Medellín Castillo ^{1,3,*} , Manuel Sánchez Polo ² , Amado Enrique Navarro Frómata ⁴  and Javier Ernesto Vilasó Cadre ⁵ 

¹ Multidisciplinary Postgraduate Program in Environmental Sciences, Av. Manuel Nava 201, 2nd. Floor, University Zone, San Luis Potosí 78000, Mexico

² Faculty of Sciences, University of Granada, 18071 Granada, Spain; mansanch@ugr.es

³ Center for Research and Postgraduate Studies, Faculty of Engineering, Autonomous University of San Luis Potosí, Dr. Manuel Nava No. 8, West University Zone, San Luis Potosí 78290, Mexico

⁴ Food and Environmental Technology Department, Technological University of Izúcar de Matamoros, De Reforma 168, Campestre La Paz, Izúcar de Matamoros 74420, Mexico; navarro4899@gmail.com

⁵ Institute of Metallurgy, Autonomous University of San Luis Potosí, Sierra Leona Av. 550, Lomas 2nd Section, San Luis Potosí 78210, Mexico; a321418@alumnos.uaslp.mx

* Correspondence: lazaroadrian1995@gmail.com (L.A.G.F.); nahum.medellin@uaslp.mx (N.A.M.C.)

Abstract: Water pollution from industrial, municipal, and agricultural sources is a pressing global concern, necessitating the development of sustainable and efficient treatment solutions. Algal biomass has emerged as a promising feedstock for the production of carbonaceous adsorbents due to its rapid growth, high photosynthetic efficiency, and ability to thrive in wastewater. This review examines the conversion of algal biomass into biochar and hydrochar through pyrolysis and hydrothermal processes, respectively, and evaluates their potential applications in wastewater treatment, carbon sequestration, and biofuel production. Pyrolyzed algal biochars typically exhibit a moderate to high carbon content and a porous structure but require activation treatments (e.g., KOH or ZnCl₂) to enhance their surface area and adsorption capabilities. Hydrothermal carbonization, conducted at lower temperatures (180–260 °C), produces hydrochars rich in oxygenated functional groups with enhanced cation exchange capacities, making them effective for pollutant removal. Algal-derived biochars and hydrochars have been successfully applied for the adsorption of heavy metals, dyes, and pharmaceutical contaminants, with adsorption capacities significantly increasing through post-treatment modifications. Beyond wastewater treatment, algal biochars serve as effective carbon sequestration materials due to their stable structure and high carbon retention. Their application as soil amendments enhances long-term carbon storage and improves soil fertility. Additionally, algal biomass plays a key role in biofuel production, particularly for biodiesel synthesis, where microalgae's high lipid content facilitates bio-oil generation. Hydrochars, with energy values in the range of 20–26 MJ/kg, are viable solid fuels for combustion and co-firing, supporting renewable energy generation. Furthermore, the integration of these materials into bioenergy systems allows for waste valorization, pollution control, and energy recovery, contributing to a sustainable circular economy. This review provides a comprehensive analysis of algal-derived biochars and hydrochars, emphasizing their physicochemical properties, adsorption performance, and post-treatment modifications. It explores their feasibility for large-scale wastewater remediation, carbon capture, and bioenergy applications, addressing current challenges and future research directions. By advancing the understanding of algal biomass as a multifunctional resource, this study highlights its potential for environmental sustainability and energy innovation.



Academic Editors: Pablo R. Salgado, Germán Giacomán-Vallejos and Adolfo Henríquez

Received: 3 February 2025

Revised: 13 February 2025

Accepted: 14 February 2025

Published: 16 February 2025

Citation: González Fernández, L.A.; Medellín Castillo, N.A.; Sánchez Polo, M.; Navarro Frómata, A.E.; Vilasó Cadre, J.E. Algal-Based Carbonaceous Materials for Environmental Remediation: Advances in Wastewater Treatment, Carbon Sequestration, and Biofuel Applications. *Processes* **2025**, *13*, 556. <https://doi.org/10.3390/pr13020556>

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

Keywords: algal biochar; hydrothermal carbonization; wastewater treatment; carbon sequestration; biofuel production

1. Introduction

Water pollution from industrial, municipal, and agricultural sources remains a critical global challenge, necessitating innovative and sustainable approaches for wastewater treatment. Conventional treatment methods, such as coagulation, flocculation, ion exchange, and membrane technologies, while effective, are often hindered by high operational costs, energy consumption, and the generation of secondary pollutants [1]. Consequently, the development of cost-effective, environmentally friendly, and efficient adsorbents has become imperative. Algal biomass, characterized by rapid growth rates, high photosynthetic efficiency, and the ability to thrive on non-arable land and wastewater, emerges as a promising feedstock for the production of carbonaceous adsorbents [2]. Conventional methods, such as activated carbon adsorption, membrane filtration, and advanced oxidation processes, are widely used for removing contaminants from water. Activated carbon, for instance, is highly effective in adsorbing organic pollutants, but its high operational cost and limited reusability are key drawbacks. Membrane filtration techniques, including reverse osmosis, provide excellent removal of dissolved contaminants but often suffer from issues like membrane fouling and high energy consumption. Advanced oxidation processes, such as ozonation and UV-H₂O₂, are effective in degrading persistent contaminants but can be energy-intensive and require careful handling of reactive chemicals [3]. In contrast, biochar and hydrochar are emerging as promising alternatives in the field of water treatment. Derived from biomass via pyrolysis or hydrothermal carbonization, both materials offer high surface areas and porous structures, making them efficient adsorbents for various contaminants. Biochar and hydrochar are not only cost-effective and sustainable but also exhibit the potential for reusability and regeneration, thus addressing some of the limitations of traditional methods. This paper explores the effectiveness of these materials for water remediation, comparing them to conventional techniques in terms of performance, sustainability, and economic feasibility [4].

Among the various conversion techniques, pyrolysis stands out as a widely used thermochemical method for transforming algal biomass into biochar under oxygen-limited conditions. Algal biochar typically exhibits a porous structure, a moderate to high carbon content, and oxygenated functional groups, with pyrolysis temperatures ranging from 300 to 650 °C. However, its surface area (1–50 m²/g) is often smaller than that of terrestrial biochars, necessitating activation processes, such as KOH or ZnCl₂ treatments, to enhance adsorptive performance [5,6]. Alternatively, hydrothermal processes provide efficient pathways for wet algal biomass conversion. Hydrothermal carbonization, conducted at moderate temperatures (180–260 °C), produces hydrochar, a carbon-rich solid with high concentrations of oxygenated functional groups and enhanced cation-exchange capacities, making it particularly effective for wastewater treatment [7].

Hydrothermal liquefaction, primarily designed for biocrude oil production, also generates solid residues with potential adsorptive applications [8]. Post-treatment modifications play a crucial role in optimizing the performance of algal-derived biochars and hydrochars. Chemical activation using agents such as KOH, ZnCl₂, or acidic treatments significantly enhances surface area, porosity, and functional group diversity, which are essential for efficient pollutant adsorption. For instance, KOH activation has been reported to increase the BET surface area of algal hydrochar by up to two orders of magnitude, leading to improved removal efficiencies for heavy metals and organic dyes [9]. Additionally, algal

chars exhibit a moderate to high carbon content (10–60%) and high ash levels, which, while detrimental for fuel applications, can be advantageous for wastewater treatment due to the presence of mineral components that enhance cation-exchange capacity [10]. In practical applications, algal-derived carbonaceous materials have demonstrated efficacy in removing heavy metals such as Pb, Cd, Cu, and Cr through mechanisms such as ion exchange and surface complexation. For example, HTC-derived hydrochar from *Sargassum hornerri* has shown high efficiency in Cr removal, while biochars from *Chlorella vulgaris* exhibit significant Cd adsorption capacities. Furthermore, algal-based adsorbents have proven effective in mitigating dye pollution from the textile industry, with activated hydrochars achieving adsorption capacities of up to 714 mg/g for methylene blue (MB), facilitated by an optimized surface area and functional group composition [11]. Similarly, pharmaceutical contaminants, including antibiotics and endocrine disruptors, have been effectively removed using chemically activated algal chars, which exhibit increased nitrogen-rich functional groups that enhance affinity for pharmaceutical molecules. Beyond wastewater treatment, algal biochars and hydrochars offer additional environmental benefits, including their use as soil amendments. Their nutrient-rich composition improves soil fertility and water retention, promoting sustainable agricultural practices.

Beyond wastewater treatment, algal chars have demonstrated significant potential in carbon capture, sequestration, and storage. Due to their high carbon content and stable structure, these materials can act as long-term carbon sinks, effectively reducing atmospheric CO₂ levels. Algal biochars have been explored for their ability to adsorb CO₂ and store it in a stable form, making them valuable in climate change mitigation strategies. Their porous nature and functional group composition enhance their CO₂ uptake capacity, providing a renewable and sustainable approach to carbon sequestration. Furthermore, algal biochars can be utilized as soil amendments to promote carbon retention in agricultural lands, further contributing to long-term carbon storage and improved soil health [12].

Algal biomass also plays a crucial role in biofuel production, offering a renewable alternative to fossil fuels. Microalgae, with their high lipid content and rapid growth rates, are particularly promising candidates for biodiesel production. Thermochemical processes such as pyrolysis and hydrothermal liquefaction can convert algal biomass into bio-oil, which can be further upgraded into transportation fuels. Additionally, algal biochars derived from biofuel production residues can be repurposed for environmental applications, creating a closed-loop system that maximizes resource efficiency. The integration of algal biochars into bioenergy systems not only supports sustainable fuel production but also provides additional environmental benefits, such as waste valorization and carbon reduction [13].

Moreover, the use of algal hydrochars in bioenergy applications extends beyond liquid fuels. With energy values ranging from 20 to 26 MJ/kg, algal hydrochars can be utilized as solid fuels for direct combustion or co-firing with conventional fuels. Their high carbon content and energy density make them a viable option for renewable energy generation. Additionally, advancements in hydrochar activation techniques can enhance their properties, making them suitable for use in fuel cells, gasification, and other advanced energy applications [14]. By integrating algal-derived biochars and hydrochars into carbon capture and bioenergy frameworks, a sustainable and multifunctional approach to environmental management can be achieved.

This review aims to provide a comprehensive analysis of the potential of algal-derived biochars and hydrochars as adsorbents for wastewater treatment while evaluating their physicochemical properties and the role of post-treatment modifications in enhancing adsorption efficiency. Additionally, it explores the feasibility of integrating these materials into

large-scale wastewater remediation processes and their potential applications in bioenergy generation, carbon capture, and sequestration. By critically examining the current state of research and identifying key challenges, this study seeks to advance the understanding of algal biomass as a versatile and sustainable solution for environmental remediation.

2. Bibliography Analysis

This review was carried out through a structured approach involving the collection, classification, analysis, and synthesis of information following the PRISMA guidelines [15]. The process commenced with comprehensive literature searches across prominent academic databases, such as Web of Science, Scopus, and CNKI, with a focus on peer-reviewed journal articles to compile the most recent advancements in algae-based technologies for environmental applications. After a thorough screening of abstracts based on predefined inclusion criteria (relevance to algae-based environmental applications, peer-reviewed journal articles, publication date, availability of full text, language, scientific quality, and exclusion of redundant studies), 1245 papers were selected for an in-depth full-text review. Ultimately, approximately 190 studies were incorporated into this analysis. Furthermore, additional data were gathered from publicly accessible sources, including reports from international organizations, governmental publications, and corporate websites. The reviewed literature spans several decades, covering research from the 1970s to 2024. By critically evaluating existing algae-based technologies for environmental applications, this review highlights current limitations and challenges while suggesting potential solutions and future research directions for further development.

3. Algae and Algal-Based Materials

Algae are a diverse group of photosynthetic organisms commonly found in aquatic environments. They exhibit a wide range of structural complexities, from unicellular microscopic forms to large multicellular species. As a polyphyletic group, algae include both prokaryotic species, such as cyanobacteria (blue-green algae), and eukaryotic species like green algae (*Chlorophyceae*), diatoms (*Bacillariophyceae*), and golden algae (*Chrysophyceae*) [16]. These organisms play a fundamental role in global ecosystems by contributing to primary production, supporting food chains, and maintaining biodiversity [17].

Algae are composed of essential biological compounds, including proteins, lipids, carbohydrates, cellulose, vitamins, minerals, and unique bioactive substances such as polysaccharides and pigments. The precise composition varies among species, influencing their applications in various industries. Key biochemical components—lipids, carbohydrates, and proteins—are particularly valuable for producing biofuels, pharmaceuticals, cosmetics, and food products [8,18].

Algae are generally categorized into two main groups based on their size and structural characteristics: macroalgae and microalgae (Figure 1). Macroalgae, often referred to as seaweeds, are large, multicellular organisms commonly found in marine environments, where they attach to surfaces such as rocks and coral reefs. Although their growth rate is relatively slow, their substantial biomass contributes significantly to carbon sequestration [19].

In contrast, microalgae are microscopic, single-celled organisms that thrive in a wide range of aquatic environments, including freshwater, brackish water, and marine systems. These organisms exhibit high photosynthetic efficiency and rapid reproduction, making them well-suited for large-scale biomass production [20]. Despite their structural differences, both macroalgae and microalgae play critical roles in carbon capture, utilization, and storage, making them valuable components of sustainable environmental and bioenergy solutions.

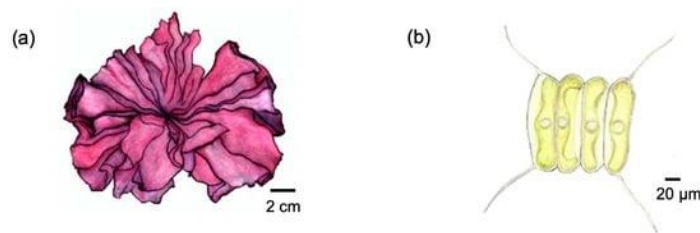


Figure 1. Diagram of algal structures: (a) *Porphyra umbilicalis* (macroalgae); (b) *Scenedesmus* (microalgae). Reproduced from Pereira, 2021 [21], under terms of Creative Commons Attribution (CC BY) license.

Environmental conditions exert a significant influence on the biochemical composition of algae, thereby affecting the characteristics of the carbon materials derived from them. Light intensity, for instance, plays a critical role in photosynthetic efficiency and carbon fixation. Under optimal light conditions, algae efficiently convert inorganic carbon into organic molecules, resulting in a biomass rich in structural carbohydrates and other carbon-based compounds. However, excessive light exposure can induce photo-oxidative stress, prompting the algae to synthesize protective compounds such as carotenoids and specific polysaccharides. These adaptations not only mitigate cellular damage but also modify the overall carbon profile of the algal biomass [22].

Temperature is another crucial factor that affects metabolic rates and enzymatic activities in algae. Moderate increases in temperature generally enhance metabolic processes and promote efficient carbon assimilation, while extreme temperatures may lead to stress responses that shift metabolic pathways. Under such stress, algae often redirect carbon flux from growth-related processes to the accumulation of storage compounds, including lipids and carbohydrates. This shift in metabolic balance can alter both the quality and the quantity of carbon materials available for downstream applications [23]. Nutrient availability further modulates the carbon composition; nutrient-limited conditions typically result in an elevated carbon-to-nitrogen ratio as algae divert fixed carbon into storage molecules rather than growth. Elevated CO_2 concentrations also enhance photosynthetic carbon fixation, potentially increasing the proportion of carbon incorporated into structural and storage compounds [24].

Additionally, variations in pH and salinity introduce osmotic and ionic stresses that influence cell wall composition and the synthesis of compatible solutes. Such environmental stresses lead to modifications in the types and arrangements of carbon-rich biomolecules within the cells, ultimately impacting the structural properties and reactivity of the derived carbon materials. The cumulative effects of these environmental parameters are especially important when algal biomass is employed as a feedstock for biofuels, biochar, or activated carbon production, where the nature of the precursor material directly determines the performance characteristics of the final product [22].

Future global conditions, particularly those driven by climate change, are poised to significantly influence the production and exploitation of new carbon materials. Climate change, characterized by rising global temperatures, altered precipitation patterns, and increased atmospheric CO_2 concentrations, directly impacts the biomass sources used as precursors in carbon material production. For instance, variations in these environmental parameters may affect the growth rates and biochemical composition of algae, a critical feedstock, thereby altering both the efficiency of carbon fixation and the quality of the resulting carbon materials [2].

In addition to these direct impacts on biomass productivity, the increased frequency and intensity of extreme weather events—such as droughts, floods, and storms—pose considerable challenges to the stability of biomass production systems. Such events can

disrupt the availability of essential resources, including water and nutrients, which are vital for consistent and high-quality biomass cultivation. Consequently, these fluctuations can lead to variability in the raw materials used for carbon synthesis, potentially affecting the efficiency, scalability, and economic viability of production processes. Moreover, the broader implications of climate change are expected to drive significant shifts in environmental policies and regulatory frameworks. As governments and international bodies implement more stringent measures to mitigate climate change, industries involved in the production of carbon materials may be compelled to adopt more sustainable practices and technologies. This regulatory evolution is likely to incentivize investments in research and development aimed at enhancing process efficiency, reducing environmental footprints, and integrating renewable energy sources into production chains [25].

Given these multifaceted challenges and opportunities, it is imperative to adopt a multidisciplinary approach that considers both the direct environmental impacts on biomass precursors and the indirect effects mediated through policy and regulatory changes. Future research should focus on developing adaptive strategies and robust process models that integrate climate projections with technological innovations. Such an approach will be essential for ensuring that the production and exploitation of new carbon materials remain resilient and sustainable in the context of evolving global environmental conditions.

4. Algal-Based Hydrochars and Biochars for Pollutants Uptake

Over time, water used in industrial processes that becomes contaminated with heavy metals has emerged as a critical environmental and public health issue. The rapid pace of industrialization and urban growth has led to the persistent release of hazardous substances—including heavy metals, synthetic dyes, pesticides, and various organic compounds—into river systems [26]. These contaminants primarily originate from wastewater produced by industries such as textiles, electroplating, paper manufacturing, leather tanning, and food processing [27]. Given their toxic and non-biodegradable nature, these pollutants tend to accumulate in the tissues of flora, fauna, and humans, leading to severe health consequences such as central nervous system damage, dermatitis, cancer, kidney failure, and impairments to liver and reproductive functions [2].

As a result, it is essential to devise efficient and cost-effective methods for the removal of heavy metals and other contaminants from industrial wastewater before its discharge into natural water bodies. In an effort to overcome the limitations of traditional purification techniques, recent research has increasingly explored the use of novel sorbents derived from renewable resources, such as hydrochars [28,29]. Hydrochar is a carbon-rich material produced through the HTC process, which involves the conversion of wet biomass into a solid, charcoal-like substance under high temperature and pressure, typically in the presence of water. Unlike biochar, which is produced through pyrolysis in the absence of oxygen, hydrochar is generated in an aqueous environment, making it particularly suitable for processing moist feedstocks such as food waste, agricultural residues, and algae. The HTC process typically occurs at temperatures ranging from 180 to 250 °C and pressures between 2 and 6 MPa [30]. Although hydrochars typically exhibit relatively low surface area and porosity, the presence of chemically active functional groups—namely ketone, carboxyl (COOH), and hydroxyl groups—on their surfaces endows them with adequate adsorption capabilities [31]. Numerous studies have investigated the potential of hydrochars produced from various precursors to adsorb heavy metals [32–34], dyes [11,35–37], and pharmaceuticals [38–41] from aqueous solutions. Moreover, a range of modification techniques have been applied to further enhance their adsorption capacity toward specific pollutants. Figure 2 outlines the potential binding mechanisms involved.

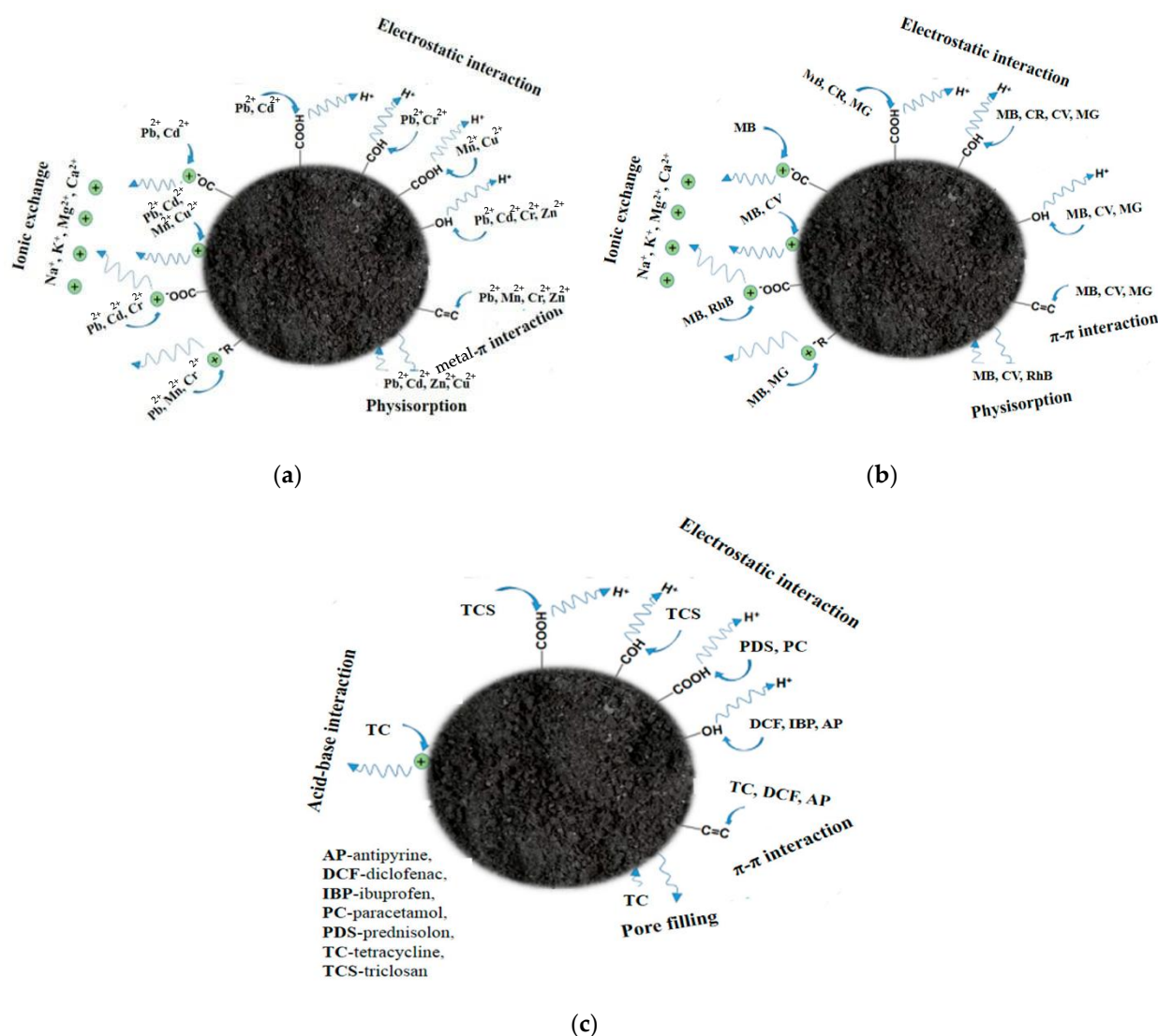


Figure 2. Possible mechanisms of interaction between hydrochar surfaces and (a) heavy metals, (b) dyes, and (c) pharmaceuticals. Images reused from Petrović et al., 2024 [42], in accordance with Creative Commons Attribution license (CC BY).

Algal biomass has gained increasing attention as a renewable feedstock for the development of sustainable carbonaceous materials. Among the various methods employed, HTC has emerged as a versatile and eco-friendly technique for converting algae into hydrochars with a high carbon content, a large surface area, and functional versatility. This section delves into studies on hydrochars derived from *Sargassum horneri*, *Sargassum muticum*, and related algal sources, focusing on their synthesis, physicochemical properties, and environmental applications (see Table 1).

A recent study by Rasam et al., 2021 [29], explored the HTC of *Sargassum horneri* under varying conditions of temperature (180–210 °C), time (2–16 h), and catalyst loading (citric acid). The hydrochar yield and properties were evaluated, including the Brunauer–Emmett–Teller (BET) surface area, higher heating value (HHV), and energy recovery. The BET surface area of the hydrochars varied significantly, ranging from 0.6 to 31.8 m²/g. The HHV ranged from 20.8 to 25.1 MJ/kg, highlighting the potential of *S. horneri*-derived hydrochar as a solid fuel. The energy densification ratios increased with reaction severity, indicating effective dehydration and decarboxylation during HTC. The optimization of HTC parameters was achieved using machine learning models such as support vector machines and artificial

neural networks. These predictive tools enhanced the understanding of complex parameter interdependencies, enabling the design of hydrochars with tailored properties for specific applications. For instance, small particle sizes and higher catalyst loads were associated with higher BET surface areas and energy recovery efficiencies, demonstrating the role of feedstock preparation and process intensification.

Table 1. Physicochemical properties and adsorptive applications of algal-derived carbonaceous materials.

Adsorbent/Feedstock	Conditions	Surface Area (m ² /g)	HHV (MJ/kg)	Adsorption Capacity (mg/g)	Key Applications	Refs.
<i>Sargassum horneri</i>	180–210 °C, 2–16 h, citric acid	0.6–31.8	20.8–25.1	-	Solid fuel, energy recovery	[29]
<i>Sargassum muticum</i>	180–300 °C, 60–300 min	22.8–24.2	-	90% (Rhodamine B)	Dye adsorption	[11]
Sargassum-derived SAH	180–260 °C, KOH activation	1216–1404	-	714.29 (MB)	Advanced adsorption	[10]
Lignocellulosic biomass	200–300 °C, HTC and activation	800–1200	22.0–26.5	600 (MB)	Adsorption and catalysis	[6]
Poultry litter hydrochar	150–300 °C, HTC	10–50	15.0–22.0	80–150 (various dyes)	Wastewater treatment	[43]
<i>Enteromorpha prolifera</i>	800 °C pyrolysis, ZnCl ₂ /FeCl ₃ activation	-	-	90 (Naphthalene, NAP), 51 (Acetaminophen, ACE), 86 (Phenanthrene, PHE)	PAH removal	[44]
<i>Aegagropila linnaei</i>	700 °C pyrolysis, KOH/Fe ₃ O ₄ loading	-	-	69.8% (Bisphenol A, BPA)	Organic pollutant removal	[45]
Microalgae-derived biochar	450–600 °C pyrolysis	-	-	218.4 and 744 (Sulfamethoxazole, STZ)	Pharmaceutical waste removal	[46,47]
Macroalgae-based biochar	800 °C pyrolysis, KOH activation	-	-	841.64 (MB)	Dye removal	[48]
<i>Nannochloropsis</i> sp.	250 °C HTC, ZnCl ₂ activation	12.56	-	34.15 (Cu)	Heavy metal removal	[49]
<i>Spirulina platensis</i>	450 °C pyrolysis	7.65	-	128.6 (MB)	Dye adsorption	[50]
<i>Cladophora glomerata</i>	350 °C pyrolysis	-	-	104 (MB)	Dye adsorption	[51]
<i>Eucheuma spinosum</i>	600 °C pyrolysis	200.74	-	331.97 (RR-120 azo dye)	Textile dye removal	[52]
<i>Chlorella vulgaris</i>	250 °C HTC	8.5	-	24.39 (Cd)	Heavy metal removal	[32]
<i>Sargassum horneri</i>	523 K HTC	14.12	-	330.84 (Cr)	Heavy metal adsorption	[53]

Another study performed by Spagnuolo et al., 2023 [11], investigated hydrochars derived from *Sargassum muticum*, emphasizing their application in removing Rhodamine B dye from wastewater. Hydrochars were synthesized under HTC conditions ranging from 180 to 300 °C for durations of 60 to 300 min. Characterization using FTIR, XPS, and SEM-EDX revealed the presence of macroporous structures and oxygen-containing functional groups that enhanced adsorption capabilities. Hydrochars obtained at lower temperatures (≤ 180 °C) and shorter durations (≤ 60 min) showed the highest adsorption capacities for Rhodamine B, attributed to their higher surface area and optimal functional group composition. Adsorption tests demonstrated removal efficiencies exceeding 90% for Rhodamine B, even without chemical or physical activation. The adsorption kinetics followed a pseudo-second-order model, indicating chemisorption as the dominant mechanism. Chambers et al., 2024 [10], examined the production of ultraporous superactivated hydrochar (SAH) from *Sargassum* using HTC followed by KOH activation. The HTC process was conducted at temperatures of 180, 220, and 260 °C, while activation enhanced the surface area, reaching up to 1404 m²/g. The resulting SAHs demonstrated remarkable adsorption capacities for MB, reaching up to 714.29 mg/g at 37 °C.

This study identified key factors influencing adsorption, including solution pH, contact time, and temperature. The optimal adsorption occurred at pH 12, with equilibrium reached within 24 h. Mechanistic insights revealed that pore filling and electrostatic attractions played critical roles in the adsorption process. The Langmuir isotherm model provided the best fit for the data, affirming monolayer adsorption behavior. Thermodynamic studies confirmed the spontaneity and endothermic nature of the adsorption process, with increased temperatures enhancing dye uptake. Comparing these studies highlights the versatility of HTC-derived hydrochars in addressing environmental challenges. While *S. horneri*-based hydrochars excel in energy recovery and fuel applications, *S. muticum*-derived hydrochars showcase high efficiency in dye removal from aqueous solutions. The addition of KOH activation further transforms hydrochars into ultraporous materials suitable for advanced adsorption applications. These findings underscore the potential of algal biomass valorization to mitigate water pollution and reduce reliance on fossil-derived adsorbents.

To promote environmental sustainability, Kim et al., 2023 [54], investigated the simultaneous removal of Cu(II) and Cr(VI) ions using a nitrogen-doped hydrochar synthesized from corncob biomass. The hydrochar was chemically modified with NH_4Cl , which significantly enhanced its adsorption efficiency, achieving capacities of 1.223 mmol/g for Cu(II) and 1.995 mmol/g for Cr(VI), surpassing those of the unmodified hydrochar. An infrared spectroscopy analysis indicated that redox reactions on the hydrochar surface led to an increase in deprotonated imine groups, which contributed to additional Cu(II) binding sites. Furthermore, under acidic conditions ($\text{pH} < \text{pH}_{\text{pzc}}$), the N-doped hydrochar exhibited a stronger affinity for Cr(VI) than Cu(II) due to electrostatic attraction. Additionally, a protonated amino-functionalized bamboo hydrochar was synthesized by modifying bamboo-derived hydrochar with acryloyl chloride, amine, and hydrochloric acid, and subsequently used for Cr(VI) adsorption [55]. The researchers highlighted that the introduction of amine groups played a crucial role in enhancing adsorption performance, achieving a capacity of 523.57 mg/g through electrostatic interactions that facilitated pollutant removal.

A novel, cost-effective hydrochar composite synthesized from sawdust and modified with magnesium and silicon (MgSi-HC) was employed for Cu(II) and Zn(II) removal. This modification resulted in an increased specific area and a highly porous structure, significantly improving its adsorption potential. Adsorption isotherm studies determined the maximum adsorption capacities for Cu(II) and Zn(II) as 214.7 mg/g and 227.3 mg/g, respectively [47]. The adsorption mechanism involved multiple interactions, including electrostatic forces, hydrogen bonding, π - π interactions, and pore filling. Furthermore, a calcium-enriched pyro-hydrochar derived from spent mushroom substrate was evaluated for its ability to capture Pb(II) and Cd(II) from aqueous solutions. Based on the Freundlich isotherm model, the maximum sorption capacities were 297 mg/g for Pb(II) and 131 mg/g for Cd(II) [56]. The removal process was attributed to several mechanisms, including ion exchange, surface complexation, mineral precipitation, and cation- π interactions.

The studies reviewed here highlight significant advancements in the modification of hydrochar-based sorbents for heavy metal removal. These materials have shown remarkable adsorption capacities and tunable surface chemistries, making them strong candidates for scalable water purification technologies. However, despite these promising developments, several research gaps remain unaddressed. One of the most pressing challenges is the limited real-world applicability of these hydrochar-based adsorbents. Most adsorption studies are conducted in controlled laboratory environments with synthetic wastewater, which does not accurately reflect the complexity of real industrial effluents. Future studies should focus on evaluating these hydrochars under realistic conditions, including multi-component wastewater systems where competition among ions might reduce adsorption

efficiency. Another crucial gap lies in the long-term stability and regeneration capacity of modified hydrochars. While adsorption efficiencies are often reported, limited studies have assessed the desorption and reusability potential of these materials. Investigating cost-effective and environmentally friendly regeneration techniques is essential to making hydrochar-based sorbents viable for large-scale use.

Additionally, mechanistic insights into adsorption processes require further exploration. While electrostatic interactions, hydrogen bonding, and π - π interactions have been proposed as key adsorption mechanisms, advanced characterization techniques such as X-ray photoelectron spectroscopy (XPS) and in situ spectroscopy should be more widely used to provide a molecular-level understanding of metal binding on hydrochar surfaces. Another area of potential advancement is the integration of hydrochar-based materials into engineered water treatment systems. Rather than using hydrochars as standalone sorbents, incorporating them into hybrid filtration systems or coupling them with membrane technologies could enhance their effectiveness and sustainability. Research in this direction could improve metal ion capture rates while ensuring the materials remain structurally intact over extended operational cycles.

Lastly, while hydrochar modifications using nitrogen, magnesium, silicon, and calcium have shown great promise, exploring alternative modification strategies—such as bio-inspired functionalization or nano-enhanced composites—could unlock even greater adsorption efficiencies. In particular, the use of biopolymer coatings or graphene-based hybrids may improve surface chemistry and expand the range of pollutants that hydrochars can target. As can be observed, hydrochar-based sorbents offer a sustainable and cost-effective approach for heavy metal removal, but their full potential remains untapped. Addressing the current limitations through real-world testing, improved regeneration strategies, deeper mechanistic studies, and advanced material integration will be critical to transitioning these adsorbents from the laboratory to large-scale environmental applications. Algal-derived hydrochars offer a sustainable approach to environmental remediation and energy recovery. Advancements in HTC and post-treatment processes, such as activation, enable the tailoring of hydrochar properties to meet specific application demands. Future research should focus on scaling up production, exploring hybrid processes, and broadening the range of pollutants addressed by these versatile materials.

Recent studies in the literature explore the applications and advancements of algae-based biochar (ABB) in environmental remediation, emphasizing its role in water treatment, air pollution control, and soil restoration. ABB is a specific type of biochar produced from algae biomass through pyrolysis, a thermochemical process that involves heating the algae in the absence of oxygen. Algae, being rich in carbon and other organic compounds, undergoes decomposition at temperatures typically ranging from 300 to 700 °C, resulting in a porous, carbon-rich material. The properties of algae-based biochar can vary depending on the species of algae used, as well as the pyrolysis conditions [46]. Algae, owing to their rapid growth, ability to thrive in adverse environments, and minimal resource demands, serve as an excellent feedstock for biochar production [57]. ABB emerges as a sustainable solution for addressing global environmental challenges due to its versatility, low cost, and renewable nature [58]. Biochar, produced through pyrolysis, gasification, and other thermal processes, is renowned for its high porosity, large specific surface area, and exceptional adsorption capacities [59] (see Table 1). Algae-based biochars, derived from both microalgae and macroalgae, outperform terrestrial plant-based biochars in several applications, primarily because of the unique composition of algae. Rich in proteins, lipids, and carbohydrates, algae contribute to the creation of biochars with enhanced functional groups and superior pollutant adsorption capabilities [60].

In water treatment, ABB has proven highly effective in removing a range of pollutants. For instance, it demonstrates remarkable efficiency in adsorbing persistent organic pollutants like polycyclic aromatic hydrocarbons (PAHs) [61,62], endocrine-disrupting chemicals (EDCs) [45], and pharmaceutical residues [47,63,64]. Studies show that *Enteromorpha prolifera*-derived biochar, activated with ZnCl_2 and FeCl_3 , exhibited adsorption capacities of 90 mg/g for NAP, 51 mg/g for ACE, and 86 mg/g for PHE [44], showcasing its potential in wastewater treatment. Similarly, algae-based biochars have been employed in the adsorption of antibiotics such as tetracycline [65] and ciprofloxacin [64], with adsorption capacities reaching up to 384.6 mg/g when biochars were modified with activators like ZnCl_2 . Furthermore, ABB is widely used in dye removal, a significant challenge posed by the textile industry. Certain macroalgae-based biochars demonstrated exceptional dye adsorption capacities, with malachite green removal reaching 5306.2 mg/g after pyrolysis at 800 °C [66]. These results underline ABB's adaptability in treating various industrial wastewater pollutants.

In the realm of soil remediation, ABB plays a dual role in reducing contamination and enhancing soil fertility [67]. Heavy metal contamination, a persistent issue in industrial and agricultural zones, is effectively mitigated using ABB. For instance, biochars derived from blue algae immobilized cadmium with an adsorption capacity of 135.7 mg/g [68]. In addition to heavy metal remediation, ABB has shown promise in reducing pesticide residues and fluoride contamination in soils. Hydroxyapatite-activated biochars have achieved defluorination efficiencies of 79% [69], while biochars tailored for pesticide residue adsorption have reduced environmental availability by over 80% [57]. Beyond pollutant mitigation, ABB significantly improves soil properties by enhancing nutrient retention [70], cation exchange capacity [71], and water-holding abilities [72]. When incorporated into agricultural practices, such as soybean cultivation, ABB has been observed to boost yields by 119% [73].

ABB also demonstrates notable performance in air pollution control, particularly in greenhouse gas adsorption and toxic gas removal [74]. For example, biochars doped with nitrogen exhibit enhanced CO_2 adsorption capacities, reaching up to 4.21 mmol/g through advanced activation methods like microwave-assisted KOH modification [75]. Similarly, ABB has been utilized in the capture of hydrogen sulfide (H_2S), a toxic gas commonly found in industrial emissions, with adsorption capacities exceeding 8500 mg/kg when activated with $\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$ [76]. The ability to modify ABB for specific pollutants makes it a versatile tool in combating air pollution, particularly for challenging contaminants like mercury, where halide-modified biochars have demonstrated removal efficiencies of over 90% [77].

Some studies thoroughly examine the various activation and modification techniques employed to enhance ABB's properties. Chemical activations, such as acid or alkali treatments, improve porosity and introduce functional groups that boost adsorption performance. For instance, KOH-modified biochars exhibited a 127% increase in CO_2 adsorption capacity [78]. Halide modifications with agents like FeCl_3 and ZnCl_2 further enhance the removal efficiency of heavy metals and organic pollutants [79,80]. Meanwhile, the integration of metal oxides like Fe_2O_3 into biochar structures not only improves catalytic properties but also facilitates the removal of complex contaminants like uranium [81]. Among the more sustainable approaches, UV/ H_2O_2 advanced oxidation processes provide a green alternative to conventional modification methods, significantly enhancing ABB's performance without contributing to secondary pollution.

Despite its potential, ABB faces challenges that hinder its widespread application. Many ABBs suffer from underdeveloped pore structures and limited active sites, which reduce their adsorption capacities. Furthermore, preparation processes can sometimes

release secondary pollutants, necessitating a careful evaluation of ABB's lifecycle impact, including its carbon footprint and energy consumption. Economic feasibility also poses a significant challenge, as scaling ABB production to industrial levels while maintaining performance enhancements can be cost-prohibitive [46].

To address these limitations, future research should focus on developing multifunctional biochars with co-doped elements and multidimensional structures to broaden their applicability. Additionally, synergistic activation techniques combining physical and chemical methods could overcome structural deficiencies. Comprehensive lifecycle assessments are essential to evaluate ABB's long-term environmental and economic impacts. Lastly, transitioning from laboratory studies to field-scale applications will be crucial in validating ABB's effectiveness and scalability under real-world conditions. As a partial conclusion, algae-based biochar represents a transformative material in environmental remediation, offering sustainable solutions for water, soil, and air pollution. By addressing its current challenges through innovative modification strategies and rigorous assessments, ABB holds the potential to become a cornerstone of sustainable environmental management.

The adsorption capacities of various macroalgae and microalgae-derived biochars for heavy metal removal from aqueous solutions have been extensively investigated under different preparation conditions. Among the macroalgae, *Macrocystis pyrifera* exhibited a significant Cu^{2+} adsorption capacity of 119.9 mg/g when subjected to pyrolysis at 450 °C under a N_2 atmosphere for one hour, with intra-particle diffusion governing the adsorption kinetics [82]. Similarly, red macroalgae treated via hydrothermal carbonization at 200 °C with FeCl_3 impregnation achieved an As^{3+} adsorption capacity of 3.8314 mg/g, following a Langmuir isotherm model and pseudo-second-order kinetics [83].

Enteromorpha prolifera demonstrated notable Cd^{2+} adsorption properties under different preparation methods. When pyrolyzed at 500 °C with KOH activation at 700 °C, the resulting biochar had a Cd^{2+} adsorption capacity of 5.84 mg/g, following the Langmuir isotherm and pseudo-second-order kinetics [84]. However, when pretreated with H_3PO_4 for 12 h before undergoing pyrolysis at 500 °C, the adsorption capacity significantly increased to 423 mg/g, with intra-particle diffusion playing a crucial role [85]. Meanwhile, *Ascophyllum nodosum* biochar, derived from pyrolysis at 700 °C followed by FeCl_3 and FeCl_2 coprecipitation, achieved a Cu^{2+} adsorption capacity of 53.19 mg/g, with pseudo-second-order kinetics dominating the adsorption process [86]. Among microalgae, *Chlorella pyrenoidosa* showed a remarkably high Ni^{2+} adsorption capacity of 201.18 mg/g when pyrolyzed at 600 °C with KOH activation, following the Langmuir isotherm and pseudo-second-order kinetics [87]. *Scenedesmus dimorphus* exhibited a lower Co^{2+} adsorption capacity of 0.672 mg/g when subjected to fast pyrolysis at 500 °C, following the Freundlich isotherm and pseudo-second-order kinetics [88].

Other notable microalgae include *Spirulina* sp., which demonstrated Cu^{2+} adsorption after pyrolysis at 200 °C, with adsorption behavior aligning with the Redlich–Peterson isotherm and pseudo-second-order kinetics [89]. Additionally, *Chlorella* sp. and *Spirulina* sp. biochars pyrolyzed at 600 °C exhibited Pb^{2+} adsorption capacities of 131.41 mg/g and 154.51 mg/g, respectively, both following Langmuir isotherm models and pseudo-second-order kinetics [90]. These findings underscore the potential of algae-derived biochars for heavy metal removal, with adsorption efficiency being highly dependent on biomass type, preparation conditions, and activation methods.

While algae-derived biochars and hydrochars show great promise for heavy metal removal from aqueous environments, several challenges and research gaps must be addressed to optimize their practical application. One of the primary concerns is the variability in the adsorption capacity of these carbon materials, which depends on the algal species used as feedstock and the preparation methods employed. As evident from previous

studies, factors such as the pyrolysis temperature, activation techniques, and chemical modifications significantly influence adsorption performance. However, there is a lack of standardized protocols for optimizing these parameters, making it difficult to compare results across studies and scale up the technology for real-world applications.

Additionally, while some biochars exhibit impressive adsorption capacities, others show relatively low efficiency, particularly for metals such as Co^{2+} and As^{3+} . This raises concerns about the selectivity of these materials and their ability to remove multiple contaminants simultaneously. Future research should focus on enhancing the functionalization of biochars to improve selectivity and binding affinity toward diverse pollutants. Advanced surface modification techniques, such as doping with nanomaterials or introducing functional groups, could be explored further. Another critical limitation is the regeneration and reusability of algal biochars. While adsorption studies highlight their high initial efficiency, there is limited research on their long-term stability and desorption performance. Repeated adsorption–desorption cycles could lead to structural degradation or reduced adsorption capacity, which may affect economic feasibility for large-scale applications [46]. Investigating effective regeneration strategies, such as thermal reactivation or chemical treatments, is crucial to ensure sustainability and cost-effectiveness.

Moreover, the economic and environmental implications of large-scale biochar production require further evaluation. Algal biomass cultivation and harvesting can be resource-intensive, particularly when grown under controlled conditions. Additionally, the energy requirements for high-temperature pyrolysis or hydrothermal processing could offset the environmental benefits if not optimized. Life cycle assessments and techno-economic analyses should be conducted to determine the true sustainability of this approach compared to conventional wastewater treatment methods. From a practical standpoint, the real-world applicability of these adsorbents depends on their performance in complex wastewater matrices, where competing ions and organic matter may interfere with adsorption. Most studies are conducted under controlled laboratory conditions with synthetic solutions, which may not accurately represent industrial or municipal wastewater. Field-scale trials are necessary to validate laboratory findings and identify potential challenges in real-world scenarios.

5. Algal-Based Materials for Carbon Capture and Energy and Biofuel Production

The integration of algal biofuel production and carbon capture catalytic approaches represents a promising frontier in sustainable energy and environmental preservation. These photosynthetic organisms, ranging from unicellular microalgae to multicellular macroalgae, possess unique characteristics that make them particularly attractive for sustainable energy solutions [91]. The increasing global demand for renewable energy sources, coupled with the urgency to mitigate greenhouse gas emissions, has intensified research into alternative bioenergy pathways. Among these, algal-based biofuels have emerged as a promising solution due to their high photosynthetic efficiency, rapid growth rates, and ability to thrive on non-arable land [92].

Microalgae and macroalgae are photosynthetic organisms that convert solar energy and carbon dioxide (CO_2) into organic compounds through photosynthesis. Their high growth rates and superior biomass productivity make them attractive candidates for biofuel production. Unlike terrestrial energy crops, algae do not compete with food production since they can be cultivated in saline, brackish, or wastewater, thereby reducing pressure on arable land. Algae are exceptional carbon fixers, utilizing photosynthesis to absorb CO_2 from the atmosphere and convert it into organic substances, primarily lipids and carbohydrates (Figure 3). This inherent ability establishes algae as natural carbon sinks,

with their cultivation playing a pivotal role in mitigating CO₂ emissions and contributing to climate change abatement.

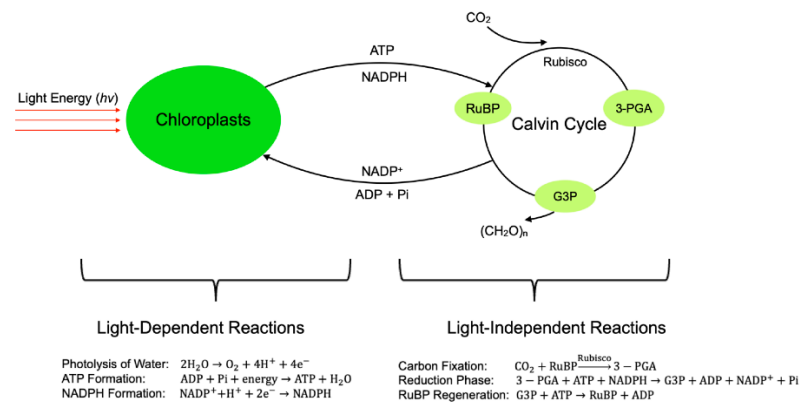


Figure 3. Photosynthetic carbon assimilation process in algae. Image reused from Li and Yao, 2024 [93], in accordance with Creative Commons Attribution license (CC BY).

The capacity of many algal species to accumulate significant amounts of lipids is a cornerstone of their utility in biodiesel production. Lipid biosynthesis in algae is intricately regulated by environmental factors such as nutrient availability, light intensity, and temperature. Under nutrient stress conditions, certain species can divert metabolic flux from growth to storage, resulting in an increased lipid content. Algal cultivation systems are broadly classified into open pond systems and closed photobioreactors (PBRs). Open ponds, due to their low operational costs, are widely used for large-scale production, albeit with limited control over environmental conditions. In contrast, PBRs offer superior control over parameters such as light, temperature, and nutrient supply, leading to higher biomass yields and more consistent product quality [94].

The cultivation of algae for biofuel production involves a range of technologies designed to maximize growth and productivity (Table 2). Open pond systems, such as shallow, sunny ponds in which algae thrive, represent a traditional and economic method. These systems are scalable and have low operating costs but may face challenges related to evaporation and contamination. Closed photobioreactors, on the other hand, offer precisely controlled environments that enable the management of temperature, light, and nutrient levels. While more expensive to set up, photobioreactors have the potential for year-round cultivation, lower contamination risks, and increased production.

Table 2. Algal processes for carbon capture and biofuel production.

Process	Description	Key Advantages	Challenges
CO ₂ capture by algae	Algae absorbs CO ₂ via photosynthesis, converting it into biomass	High CO ₂ fixation rates, renewable process	Requires optimized growth conditions and nutrient supply
Open-pond cultivation	Algae grow in large outdoor ponds under natural sunlight	Low-cost and scalable	Susceptible to contamination and low productivity
Photobioreactor cultivation	Algae cultivated in enclosed systems with controlled light and nutrients	High biomass yield, reduced contamination	High capital and operational costs
Biofuel production (transesterification)	Converts algal lipids into biodiesel using catalysts	Produces clean, renewable fuel	Energy-intensive lipid extraction and purification required
Hydrothermal liquefaction	Converts whole algal biomass into biocrude oil using high temperature and pressure	Utilizes wet biomass, high energy content fuel	High processing costs, catalyst deactivation
Pyrolysis	Thermal decomposition of algae to produce syngas and bio-oil	Efficient energy recovery	Complex upgrading processes required
Carbon storage in biochar	Converts algal biomass into stable carbon-rich biochar	Long-term carbon sequestration potential	Requires large-scale applications for impact

The efficient harvesting of algal biomass remains one of the most critical challenges in algal biofuel production. Techniques such as flocculation, centrifugation, and membrane filtration are commonly employed [95]. Emerging methods, including bio-flocculation and electrocoagulation, have shown promise in reducing energy consumption and operational costs. Thermochemical processes, such as pyrolysis, gasification, and hydrothermal liquefaction, convert algal biomass into liquid and gaseous fuels. Pyrolysis involves the thermal decomposition of biomass in the absence of oxygen, yielding bio-oil, char, and syngas. Gasification, on the other hand, converts biomass into a combustible gas mixture through partial oxidation. Hydrothermal liquefaction operates at high pressures and moderate temperatures to produce bio-crude oil, which can be further refined into transportation fuels. These processes benefit from the high moisture content of algae, which can be advantageous under certain processing conditions [96].

Biochemical conversion routes include anaerobic digestion and fermentation. The anaerobic digestion of algal biomass produces biogas, primarily composed of methane, while fermentation processes convert carbohydrates into ethanol and other bioalcohols. The biochemical conversion of algal biomass is often limited by the structural complexity of the cell wall; thus, pretreatment methods such as mechanical disruption, enzymatic hydrolysis, or acid treatments are required to improve conversion efficiency. The integration of biochemical conversion with advanced biotechnological approaches has the potential to enhance overall yields and reduce processing costs [97]. Recent research has focused on integrating thermochemical and biochemical conversion techniques to optimize biofuel production. Hybrid processes that combine hydrothermal liquefaction with subsequent catalytic upgrading have shown enhanced conversion efficiencies and improved fuel properties. These integrated approaches leverage the strengths of each conversion pathway, resulting in biofuels that meet stringent performance and emission standards [98].

Catalytic processes play a crucial role in harnessing the immense potential of algae as a biofuel source. In the context of algae-based biofuel production (Figure 4), catalysis is indispensable for several critical steps. Transesterification, a primary process in biodiesel production, relies heavily on catalysts such as sodium hydroxide or potassium hydroxide to convert algal lipids into biodiesel efficiently [99]. This transformation yields biodiesel and glycerol while ensuring reaction efficiency and the production of high-quality fuel. Hydrothermal liquefaction offers another promising route, subjecting algal biomass to high temperatures and pressures to break it down into valuable biocrude oil [100]. Catalysts are instrumental in enhancing reaction rates and yields, rendering the process economically viable. Additionally, the gasification of algal biomass generates synthesis gas (syngas) that can be further refined into liquid biofuels or valuable chemicals. Catalysts optimize the gasification process, facilitating the efficient conversion of abundant raw algal materials into refined, energy-dense biofuels suitable for diverse applications, including transportation, industry, and power generation [101].

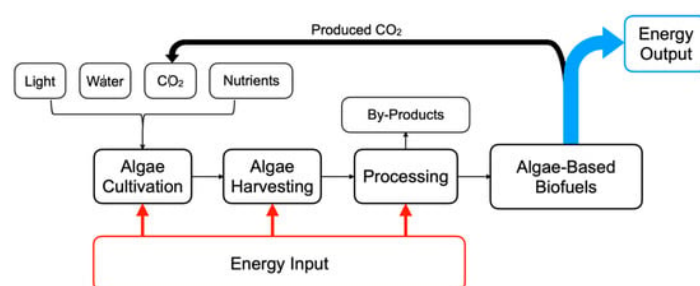


Figure 4. Biofuel generation process from algae. Image reused from Li and Yao, 2024 [93], in accordance with Creative Commons Attribution license (CC BY).

The cultivation of algae for biofuel production involves a range of technologies designed to maximize growth and productivity. Open pond systems, such as shallow, sunny ponds in which algae thrive, represent a traditional and economic method. These systems are scalable and have low operating costs but may face challenges related to evaporation and contamination [102]. Closed photobioreactors, on the other hand, offer precisely controlled environments that enable the management of temperature, light, and nutrient levels. While more expensive to set up, photobioreactors have the potential for year-round cultivation, lower contamination risks, and increased production [94]. Emerging technologies such as raceway ponds and vertical tube bioreactors combine the advantages of open ponds and photobioreactors, aiming to maximize productivity while reducing costs and operational complexity [103].

Technological advancements in algae cultivation focus on improving productivity, resource efficiency, and sustainability. Innovative approaches utilize genetic engineering to develop high-yielding algae strains that are more resilient to environmental stressors or have a higher lipid content. Integration with wastewater treatment plants enables nutrient recycling, reducing production costs and environmental impact. Furthermore, combining biomass production with carbon capture technologies enhances environmental sustainability by reducing greenhouse gas emissions.

A multitude of factors influence the cultivation of microalgae species, playing a pivotal role in determining overall productivity and economic viability. Environmental factors such as light intensity and quality, temperature, pH, and carbon dioxide concentration are critical in ensuring optimal growth conditions. Operational factors, including nutrient availability, mixing and turbulence, efficient harvesting and dewatering methods, and contamination control, are equally crucial for maintaining high biomass yields and preventing losses. Biological factors, like strain selection, genetic modifications, life cycle stage, and metabolic pathways, significantly impact the production of desired compounds and the overall performance of the cultivation system. The design of the cultivation system itself, whether open or closed, and its scale and geometry, can profoundly affect factors such as light penetration, mixing, gas exchange, and compatibility with downstream processing steps [104].

Recent innovations in genetic engineering, metabolic engineering, and synthetic biology have revolutionized the ability to enhance algal traits with greater speed and precision. These advancements provide promising avenues for improving algae's efficiency in carbon capture and biofuel production. Yang et al., 2017 [105], employed genetic engineering to enhance the photosynthetic performance of *Chlorella vulgaris*. By introducing cyanobacterial fructose-1,6-bisphosphate aldolase via a plastid transit peptide, they significantly boosted photosynthetic capacity and cell growth. Their research suggests that overexpressing this enzyme facilitates the regeneration of ribulose-1,5-bisphosphate in the Calvin cycle, improving energy transfer within photosystems and ultimately enhancing carbon fixation efficiency. Similarly, Hlavova et al., 2015 [106], conducted an extensive review of genetic engineering strategies for modifying microalgal traits. These techniques enable the precise insertion of exogenous DNA or external induction mechanisms, leading to genetically optimized strains with enhanced biomass production, increased cell density, and improved CO₂ capture efficiency.

Beyond genetic modifications, metabolic engineering has emerged as a critical tool for fine-tuning algal biochemical pathways. Brar et al., 2021 [107], highlighted strategies such as modifying enzyme expression levels, altering protein structures for increased stability, and optimizing key metabolic reactions to enhance CO₂-to-biomass conversion. These approaches have been instrumental in improving algal productivity under varying environmental conditions. Synthetic biology further expands the potential for algal biotechnology by integrating artificial genetic modifications with metabolic pathway en-

gineering. Among the most promising tools in this field, CRISPR-Cas systems have been successfully adapted to enhance algal carbon sequestration and biofuel synthesis. For instance, modifications to *Chlamydomonas reinhardtii* and *Synechocystis sp.* have improved their resilience to environmental stressors while optimizing light-harvesting pathways for accelerated growth. Additionally, combining CRISPR-based techniques with high-throughput screening holds significant potential for scaling up algae-based carbon capture and bioenergy applications [108]. By optimizing key enzymatic steps and redirecting metabolic fluxes, researchers have achieved significant improvements in biomass composition and biofuel precursor synthesis [109]. Computational models and high-throughput screening techniques are now routinely used to predict and validate genetic modifications, thus accelerating the development of high-performance algal strains. These advances are critical for overcoming bottlenecks in the conversion of CO₂ to biofuels and enhancing overall process efficiency.

Despite these promising advances, a critical gap remains in transitioning these laboratory successes to large-scale, real-world applications. While genetically engineered algae have demonstrated enhanced performance in controlled environments, their stability and efficiency in open-pond or industrial-scale bioreactors remain uncertain. Factors such as genetic drift, regulatory challenges, ecological risks, and cost-effectiveness need further exploration. Future research should focus on bridging this gap by conducting extensive field trials and developing robust biocontainment strategies to ensure the viability and safety of engineered algal strains in large-scale carbon capture, utilization, and storage (CCUS) systems. Addressing these challenges will be key to unlocking the full potential of algae as a sustainable solution for carbon mitigation and biofuel production.

Extracting lipids from microalgal biomass is also a critical step in the production of biofuels and other valuable compounds. Several extraction techniques have been developed and employed, each with its own advantages and limitations. Conventional solvent extraction methods, such as hexane extraction [110] or the Bligh and Dyer method [111], are widely used for their efficiency in extracting lipids. However, these techniques often require large volumes of toxic organic solvents, raising environmental and safety concerns. Emerging techniques like supercritical fluid extraction, using carbon dioxide or other solvents under high pressure and temperature, offer a greener and more selective approach but may require specialized equipment and higher operational costs [112].

Innovative techniques like enzymatic extraction, utilizing lipase enzymes to break down cell walls and release lipids, and ultrasound-assisted extraction, employing high-frequency sound waves to disrupt cell membranes, have gained attention for their potential to improve extraction efficiency and reduce solvent usage. Microwave-assisted extraction and pulsed electric field extraction have shown promise in enhancing the permeability of cell membranes and facilitating lipid recovery. The choice of extraction technique often depends on factors such as the microalgal species, desired lipid quality, scalability, environmental impact, and economic viability, necessitating a careful evaluation of the trade-offs between efficiency, sustainability, and cost [12].

Microalgae species such as *Chlorella*, *Nannochloropsis*, and *Scenedesmus* are frequently chosen for biodiesel production due to their high lipid content and rapid growth rates. These microalgae can accumulate triacylglycerols, which can be transesterified to produce biodiesel. *Chlorella* has been extensively studied for biodiesel production due to its high lipid content and adaptability to various environmental conditions. *Nannochloropsis* is another promising option for biodiesel feedstock because of its robust growth and lipid-rich biomass. Similarly, *Scenedesmus* has shown potential for biodiesel production due to its ability to accumulate lipids under various cultivation conditions [104].

The integration of algae-based technologies into existing energy infrastructure presents both opportunities and challenges. Understanding the synergies and trade-offs between algal biofuel production and carbon capture is crucial for maximizing their impact on climate change mitigation and sustainable development. Scalability remains a significant hurdle in translating laboratory success to large-scale industrial applications. Optimizing cultivation techniques, improving strain selection for higher lipid content, and developing efficient harvesting methods are essential steps in addressing this challenge.

To overcome challenges associated with large-scale algal cultivation, researchers have explored mixed-culture systems, where multiple algal species are cultivated together [113–117]. This approach takes advantage of species diversity, allowing cultures to better adapt to fluctuating environmental conditions such as light availability, temperature shifts, and pH variations. Additionally, mixed cultures can suppress the proliferation of pathogens and unwanted contaminants, reducing the risk of culture crashes. By fostering complementary nutrient utilization among species, these systems not only enhance biomass productivity but also lower operational costs, making large-scale cultivation more economically viable. One promising method for optimizing mixed-culture systems involves leveraging natural density differences between algal strains. This allows for the selective enrichment of lipid-rich or carbohydrate-rich species, simplifying biomass harvesting and improving the efficiency of biofuel and bioproduct generation. Such strategies provide a cost-effective alternative to traditional separation techniques, which are often energy-intensive and expensive [118].

In addition to mixed cultures, integrating algal cultivation with industrial byproducts such as wastewater and CO₂-rich emissions presents another sustainable solution. Wastewater from agricultural and food processing industries contains essential nutrients like nitrogen and phosphorus, which can serve as a growth medium for algae. By utilizing this resource, algal cultivation systems not only achieve higher biomass yields but also contribute to wastewater remediation, reducing the environmental impact of nutrient pollution. Studies have shown that microalgal strains can thrive in wastewater conditions, effectively removing excess nutrients while simultaneously increasing biomass production [119].

Similarly, the direct utilization of CO₂-rich industrial emissions has been explored as a way to enhance algal carbon capture efficiency. Positioning cultivation systems near emission sources, such as power plants or industrial facilities, minimizes the costs associated with CO₂ transportation while improving carbon fixation rates. Certain thermotolerant microalgal strains have been successfully cultivated using exhaust gases from industrial burners, demonstrating the feasibility of large-scale integration with existing infrastructure. Open raceway ponds located near power plants have also been used to capture flue gas emissions, highlighting the potential for synergy between carbon-emitting industries and sustainable algal production [120].

Despite these advancements, several challenges remain. The long-term stability of mixed-culture systems under outdoor conditions is still not fully understood, as interactions between species can shift over time, potentially leading to competitive imbalances. Additionally, while wastewater and flue gas integration offer promising cost reductions, optimizing nutrient availability and CO₂ transfer rates for consistent biomass productivity requires further investigation. Future research should focus on developing more predictive models for species interactions in mixed cultures and refining scalable methods for nutrient and carbon delivery in industrial applications. Addressing these gaps will be crucial in realizing the full potential of algae as a sustainable solution for carbon capture and biofuel production.

Moreover, the development of cost-effective and durable catalysts is necessary to enhance carbon capture efficiency and address challenges such as low stability and high

cost. Research efforts are focused on designing novel catalysts that can withstand the harsh conditions of carbon capture processes while maintaining high efficiency and selectivity. Nanomaterials and hybrid catalysts have shown promise in this regard, offering improved performance and longevity compared to traditional catalysts. The environmental benefits of algae-based catalysis extend beyond carbon capture and biofuel production. Algae-based systems can be utilized for wastewater treatment, removing nutrients and contaminants while simultaneously producing valuable biomass. This dual-purpose approach addresses water pollution issues while generating feedstock for biofuel production. Additionally, algae-derived materials have shown potential in photocatalytic and electrocatalytic applications, offering sustainable alternatives for various industrial processes.

As research in algae-based technologies progresses, addressing current challenges and exploring future directions are crucial for realizing their full potential. Improving the efficiency of lipid extraction and conversion processes, developing more robust and productive algal strains through genetic engineering, and optimizing large-scale cultivation systems are key areas of focus. Furthermore, integrating algae-based technologies with other renewable energy sources and industrial processes can create synergistic effects, enhancing overall sustainability and economic viability.

The integration of algal biofuel production and carbon capture catalytic approaches represents a promising pathway towards a more sustainable and environmentally friendly energy future. By harnessing the unique capabilities of algae and leveraging advanced catalytic technologies, these integrated systems offer solutions to multiple global challenges, including climate change mitigation, energy security, and environmental preservation. Continued research, innovation, and interdisciplinary collaboration will be essential in overcoming current limitations and unlocking the full potential of algae-based technologies. As the world transitions towards a low-carbon economy, algae-based solutions are poised to play a significant role in shaping a cleaner, more sustainable future.

6. Algae-Based Solid Biomass Pellets

Algae-based solid biomass pellets have emerged as a promising renewable fuel source, offering high energy density and efficient combustion with minimal carbon emissions. These pellets, primarily derived from microalgae, undergo densification processes that improve their storage, transport, and usability in existing combustion systems. Their compatibility with coal-fired boilers makes them a viable alternative to traditional fossil fuels while also contributing to carbon recycling [121]. The production of algae-based biomass pellets involves several key stages: drying, crushing, screening, and palletization. Drying, often achieved through solar dryers or mechanical drying equipment, is crucial for reducing moisture content, thereby increasing combustion efficiency. Once dried, the biomass is crushed and screened to ensure a uniform particle size before being compressed into dense pellets. This densification step enhances their transportability and energy output [122].

To further improve pellet performance, researchers have explored techniques such as increasing compression pressure, incorporating additives, and using higher processing temperatures. One particularly effective method is torrefaction palletization, which involves mild thermal treatment (200–300 °C) in an oxygen-limited environment. This process removes excess moisture and volatile compounds, yielding pellets with superior energy density, structural integrity, and durability [123]. A significant advantage of algae-based pellets is their clean combustion profile. When burned, they primarily release CO₂ and ash—both of which can be reintegrated into algal cultivation systems. The emitted CO₂ acts as a carbon source for photosynthesis, while ash, rich in essential minerals, serves as a nutrient supplement for further algae growth. Studies have shown that these pellets

can achieve high energy densities comparable to or exceeding conventional biomass fuels, underscoring their potential for bioenergy applications [124,125].

To address cost challenges and enhance fuel performance, researchers have explored blending algae with other biomass materials, such as wood waste, agricultural residues, and rice husks. Even in small proportions, algae contribute to improved combustion efficiency, higher energy density, and enhanced durability of the pellets. Their fine particle size allows them to fill voids in biomass mixtures, increasing pellet density, while their lipid content improves structural cohesion, reducing fragmentation during burning. Additionally, the natural oils in algae act as lubricants, lowering energy requirements during the palletization process.

Beyond standalone pellets, algae-based fuels have been tested in dual-fuel applications, where they are co-combusted with coal, sludge, or oil shale [126–129]. These combinations offer multiple advantages, such as reducing ignition temperatures, accelerating burnout rates, and enhancing overall combustion performance. Algae can also facilitate the breakdown of sludge residues and contribute to more stable combustion reactions, lowering emissions of harmful pollutants such as nitrogen oxides. While algae-based biomass pellets demonstrate significant potential, challenges remain in scaling up their production and optimizing their economic feasibility. One key research gap lies in the long-term stability and combustion behavior of algae-blended fuels in real-world industrial applications. Additionally, refining torrefaction conditions to maximize energy retention while minimizing production costs remains an ongoing area of investigation. Future studies should also focus on integrating algae-based pellets into existing energy infrastructure, assessing their performance in different combustion environments, and developing predictive models for their large-scale deployment. By addressing these challenges, algae-based biomass pellets could become a pivotal solution in sustainable energy production, reducing dependence on fossil fuels while advancing carbon-neutral bioenergy technologies.

7. Conclusions

Algal-based biochar and hydrochar have emerged as promising and sustainable materials for wastewater treatment, offering efficient removal of heavy metals, dyes, pharmaceuticals, and other pollutants. Their production through pyrolysis, HTC, or HTL allows for the development of highly tunable adsorbents, whose performance can be further enhanced through advanced activation and modification techniques. Beyond water remediation, algal-derived chars provide additional benefits, including soil amendment and energy recovery, reinforcing their potential as a versatile tool in environmental engineering. However, challenges related to feedstock variability, production costs, and large-scale implementation must still be addressed to unlock their full potential. Future research should focus on refining processing techniques, optimizing activation methods, and improving regeneration strategies to ensure long-term stability and effectiveness.

As can be observed, while algal-based biochars and hydrochars offer a sustainable and innovative approach to heavy metal, dye, and pharmaceutical removal, addressing these gaps is crucial for their widespread adoption. Future research should focus on improving selectivity, enhancing regeneration strategies, reducing production costs, and conducting real-world trials to bridge the gap between laboratory research and practical implementation. By tackling these limitations, algal-based adsorbents could become a key component of sustainable wastewater treatment technologies.

This review also highlights the broader role of algae-based CCUS technologies in mitigating climate change while contributing to sustainable energy solutions. Recent advances in algal biotechnology, including genetic modifications, optimized cultivation methods, and enhanced carbon capture efficiency, have significantly improved the conversion of

CO₂ into valuable biomass. This biomass serves as a feedstock for various biofuels—solid, liquid (biodiesel and bioethanol), and gaseous (biohydrogen, biogas, and syngas)—with technological innovations continuously increasing fuel yields. Despite these advancements, economic feasibility remains a major barrier to widespread adoption. Reducing cultivation costs, improving conversion efficiency, and minimizing energy inputs are critical areas for future development. Additionally, integrating algae-based biofuels into existing energy infrastructure and exploring biorefinery models could enhance their commercial viability.

HTC has also gained attention as an efficient thermochemical process for converting wet biomass into hydrochars, which have diverse applications in energy storage, adsorption, and soil conditioning. Understanding key process parameters—such as temperature, pressure, and residence time—has allowed better control over hydrochar properties, optimizing them for specific environmental applications. While significant progress has been made in elucidating the mechanisms of hydrothermal conversion, further research is needed to improve hydrochar performance, particularly through surface modifications and regeneration strategies. Enhancing hydrochar stability and adsorption capacity is crucial for its large-scale application in environmental remediation.

Algal-based biofuels and carbon materials present a multifaceted solution to the twin challenges of energy security and environmental sustainability. Their high photosynthetic efficiency, rapid growth rates, and diverse biochemical composition offer key advantages over conventional energy crops. However, the successful commercialization of these technologies will require a concerted effort across multiple disciplines, including genetic engineering, process optimization, and integrated system design. Moving forward, the combination of innovative research, technological advancements, and supportive policies will be essential in transforming algae-based solutions from a promising concept into a viable alternative for carbon mitigation, biofuel production, and environmental restoration.

Author Contributions: L.A.G.F.: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing—Original Draft, and Visualization. N.A.M.C.: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Data Curation, Writing—Original Draft, Visualization, Resources, and Supervision. M.S.P.: Visualization, Resources, and Supervision. A.E.N.F.: Visualization, Resources, and Supervision. J.E.V.C.: Conceptualization, Methodology, and Writing—Original Draft. All authors have read and agreed to the published version of the manuscript.

Funding: The authors thank CONAHCyT (CVU 1014829) for funding this research. This work forms part of a doctoral thesis in Chemistry at the University of Granada (Spain) and in Environmental Sciences at the Autonomous University of San Luis Potosi (Mexico).

Data Availability Statement: Data will be made available upon request.

Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

1. Briffa, J.; Sinagra, E.; Blundell, R. Heavy Metal Pollution in the Environment and Their Toxicological Effects on Humans. *Heliyon* **2020**, *6*, e04691. [[CrossRef](#)] [[PubMed](#)]
2. Ramesh, B.; Saravanan, A.; Senthil Kumar, P.; Yaashikaa, P.R.; Thamarai, P.; Shaji, A.; Rangasamy, G. A Review on Algae Biosorption for the Removal of Hazardous Pollutants from Wastewater: Limiting Factors, Prospects and Recommendations. *Environ. Pollut.* **2023**, *327*, 121572. [[CrossRef](#)]
3. Qi, Y.; He, K. Science and Technology for Water Purification: Achievements and Strategies. *Water* **2025**, *17*, 91. [[CrossRef](#)]
4. Arora, N.; Tripathi, S.; Bhatnagar, P.; Gururani, P.; Philippidis, G.P.; Kumar, V.; Mohan Poluri, K.; Nanda, M. Algal-Based Biochar and Hydrochar: A Holistic and Sustainable Approach to Wastewater Treatment. *Chem. Eng. J.* **2024**, *496*, 153953. [[CrossRef](#)]
5. Nguyen, M.H.; Zbair, M.; Dutournié, P.; Gervasini, A.; Vaulot, C.; Bennici, S. Toward New Low-Temperature Thermochemical Heat Storage Materials: Investigation of Hydration/Dehydration Behaviors of MgSO₄/Hydroxyapatite Composite. *Sol. Energy Mater. Sol. Cells* **2022**, *240*, 111696. [[CrossRef](#)]

6. Zeng, G.; Lou, S.; Ying, H.; Wu, X.; Dou, X.; Ai, N.; Wang, J. Preparation of Microporous Carbon from *Sargassum Horneri* by Hydrothermal Carbonization and KOH Activation for CO₂ Capture. *J. Chem.* **2018**, *2018*, 4319149. [\[CrossRef\]](#)
7. Magdziarz, A.; Wilk, M.; Wądrzyk, M. Pyrolysis of Hydrochar Derived from Biomass—Experimental Investigation. In Proceedings of the ECOS 2019—Proceedings of the 32nd International Conference on Efficiency, Cost, Optimization, Simulation and Environmental Impact of Energy Systems, Wroclaw, Poland, 23–28 June 2019; Volume 2019-June, pp. 4465–4476.
8. Borah, D.; Rout, J.; Nooruddin, T. Application of Nanotechnology in Bioenergy Production from Algae and Cyanobacteria. In *Modern Nanotechnology: Volume 2: Green Synthesis, Sustainable Energy and Impacts*; Springer: Berlin/Heidelberg, Germany, 2023; pp. 267–291. ISBN 9783031311048.
9. El Ouadrhiri, F.; Elyemni, M.; Lahkimi, A.; Lhassani, A.; Chaouch, M.; Taleb, M. Mesoporous Carbon from Optimized Date Stone Hydrochar by Catalytic Hydrothermal Carbonization Using Response Surface Methodology: Application to Dyes Adsorption. *Int. J. Chem. Eng.* **2021**, *2021*, 1–16. [\[CrossRef\]](#)
10. Chambers, C.; Saha, S.; Grimes, S.; Calhoun, J.; Reza, M.T. Physical and Morphological Alteration of *Sargassum*-Derived Ultraporous Superactivated Hydrochar with Remarkable Cationic Dye Adsorption. *Biomass Convers. Biorefinery* **2023**, *14*, 29131–29144. [\[CrossRef\]](#)
11. Spagnuolo, D.; Iannazzo, D.; Len, T.; Balu, A.M.; Morabito, M.; Genovese, G.; Espro, C.; Bressi, V. Hydrochar from *Sargassum Muticum*: A Sustainable Approach for High-Capacity Removal of Rhodamine B Dye. *RSC Sustain.* **2023**, *1*, 1404–1415. [\[CrossRef\]](#)
12. Sahu, S.; Kunj, P.; Kaur, A.; Khatri, M.; Singh, G.; Arya, S.K. Catalytic Strategies for Algal-Based Carbon Capture and Renewable Energy: A Review on a Sustainable Approach. *Energy Convers. Manag.* **2024**, *310*, 118467. [\[CrossRef\]](#)
13. Thanigaivel, S.; Rajendran, S.; Gnanasekaran, L.; Chew, K.W.; Tran, D.T.; Tran, H.D.; Nghia, N.K.; Show, P.L. Nanotechnology for Improved Production of Algal Biofuels: A Review. *Environ. Chem. Lett.* **2023**, *21*, 821–837. [\[CrossRef\]](#)
14. Baghel, R.S.; Suthar, P.; Gajaria, T.K.; Bhattacharya, S.; Anil, A.; Reddy, C.R.K. Seaweed Biorefinery: A Sustainable Process for Valorising the Biomass of Brown Seaweed. *J. Clean. Prod.* **2020**, *263*, 121359. [\[CrossRef\]](#)
15. Selcuk, A.A. A Guide for Systematic Reviews: PRISMA. *Turk. Arch. Otorhinolaryngol.* **2019**, *57*, 57–58. [\[CrossRef\]](#)
16. Kumar, B.R.; Mathimani, T.; Sudhakar, M.P.; Rajendran, K.; Nizami, A.S.; Brindhadevi, K.; Pugazhendhi, A. A State of the Art Review on the Cultivation of Algae for Energy and Other Valuable Products: Application, Challenges, and Opportunities. *Renew. Sustain. Energy Rev.* **2021**, *138*, 110649. [\[CrossRef\]](#)
17. Vatsha, P.; Alam, M.R. Environmental Aspects of Biofuels Used as Sustainable Energy Resources: Current Situation and Future Trends. In *Biofuels and Sustainability Life Cycle Assessments, System Biology, Policies, and Emerging Technologies*; Elsevier: Amsterdam, The Netherlands, 2024; pp. 19–31. [\[CrossRef\]](#)
18. Davis, T.A.; Volesky, B.; Mucci, A. A Review of the Biochemistry of Heavy Metal Biosorption by Brown Algae. *Water Res.* **2003**, *37*, 4311–4330. [\[CrossRef\]](#) [\[PubMed\]](#)
19. Moreda, U.; Mazarrasa, I.; Cebrian, E.; Kaal, J.; Ricart, A.M.; Serrano, E.; Serrano, O. Role of Macroalgal Forests within Mediterranean Shallow Bays in Blue Carbon Storage. *Sci. Total Environ.* **2024**, *934*, 173219. [\[CrossRef\]](#)
20. Gaurav, K.; Neeti, K.; Singh, R. Microalgae-Based Biodiesel Production and Its Challenges and Future Opportunities: A Review. *Green Technol. Sustain.* **2024**, *2*, 100060. [\[CrossRef\]](#)
21. Pereira, L. Macroalgae. *Encyclopedia* **2021**, *1*, 177–188. [\[CrossRef\]](#)
22. Hu, Q. Environmental Effects on Cell Composition. In *Handbook of Microalgal Culture: Applied Phycology and Biotechnology*, 2nd ed.; Wiley: Hoboken, NJ, USA, 2013; pp. 114–122. [\[CrossRef\]](#)
23. Davison, I.R. Environmental Effects on Algal Photosynthesis: Temperature. *J. Phycol.* **1991**, *27*, 2–8. [\[CrossRef\]](#)
24. Richmond, A. Cell Response to Environmental Factors. In *CRC Handbook of Microalgal Mass Culture*; CRC Press: Boca Raton, FL, USA, 2017; pp. 69–99. ISBN 9781351362702.
25. Stevenson, J. Ecological Assessments with Algae: A Review and Synthesis. *J. Phycol.* **2014**, *50*, 437–461. [\[CrossRef\]](#)
26. Zamora-Ledezma, C.; Negrete-Bolagay, D.; Figueroa, F.; Zamora-Ledezma, E.; Ni, M.; Alexis, F.; Guerrero, V.H. Heavy Metal Water Pollution: A Fresh Look about Hazards, Novel and Conventional Remediation Methods. *Environ. Technol. Innov.* **2021**, *22*, 101504. [\[CrossRef\]](#)
27. Akbar, S.A.; Khairunnisa, K. Seaweed-Based Biosorbent for the Removal of Organic and Inorganic Contaminants from Water: A Systematic Review. In *BIO Web of Conferences*; EDP Sciences: Les Ulis, France, 2024; Volume 87, p. 2011.
28. Periyavaram, S.R.; Bella, K.; Uppala, L.; Reddy, P.H.P. Hydrothermal Carbonization of Food Waste: Process Parameters Optimization and Biomethane Potential Evaluation of Process Water. *J. Environ. Manag.* **2023**, *347*, 119132. [\[CrossRef\]](#)
29. Rasam, S.; Talebkeikhah, F.; Talebkeikhah, M.; Salimi, A.; Moraveji, M.K. Physico-Chemical Properties Prediction of Hydrochar in Macroalgae *Sargassum Horneri* Hydrothermal Carbonisation. *Int. J. Environ. Anal. Chem.* **2021**, *101*, 2297–2318. [\[CrossRef\]](#)
30. Zhang, Z.; Zhu, Z.; Shen, B.; Liu, L. Insights into Biochar and Hydrochar Production and Applications: A Review. *Energy* **2019**, *171*, 581–598. [\[CrossRef\]](#)
31. Santos Santana, M.; Pereira Alves, R.; da Silva Borges, W.M.; Francisquini, E.; Guerreiro, M.C. Hydrochar Production from Defective Coffee Beans by Hydrothermal Carbonization. *Bioresour. Technol.* **2020**, *300*, 122653. [\[CrossRef\]](#) [\[PubMed\]](#)

32. Sufian, J.; Sari, M.B.; Marchelli, F.; Fiori, L.; Avanes, A.; Moradi, S. Algal Biomass, Biochar and Hydrochar from *Chlorella Vulgaris* for Cadmium Removal from Aqueous Streams. *Res. Sq.* **2023**. [\[CrossRef\]](#)
33. Zhang, X.; Gao, Z.; Fan, X.; Tan, L.; Jiang, Y.; Zheng, W.; Han, F.X.; Liang, Y. A Comparative Study on Adsorption of Cadmium and Lead by Hydrochars and Biochars Derived from Rice Husk and *Zizania Latifolia* Straw. *Environ. Sci. Pollut. Res.* **2022**, *29*, 63768–63781. [\[CrossRef\]](#)
34. Wang, J.; Wang, Y.; Wang, J.; Du, G.; Khan, K.Y.; Song, Y.; Cui, X.; Cheng, Z.; Yan, B.; Chen, G. Comparison of Cadmium Adsorption by Hydrochar and Pyrochar Derived from Napier Grass. *Chemosphere* **2022**, *308*, 136389. [\[CrossRef\]](#) [\[PubMed\]](#)
35. Haris, M.; Khan, M.W.; Paz-Ferreiro, J.; Mahmood, N.; Eshtiaghi, N. Synthesis of Functional Hydrochar from Olive Waste for Simultaneous Removal of Azo and Non-Azo Dyes from Water. *Chem. Eng. J. Adv.* **2022**, *9*, 100233. [\[CrossRef\]](#)
36. Kohzadi, S.; Marzban, N.; Libra, J.A.; Bundschuh, M.; Maleki, A. Removal of RhB from Water by Fe-Modified Hydrochar and Biochar—An Experimental Evaluation Supported by Genetic Programming. *J. Mol. Liq.* **2023**, *369*, 120971. [\[CrossRef\]](#)
37. Algethami, J.S.; Alhamami, M.A.M.; Alqadami, A.A.; Melhi, S.; Seliem, A.F. Magnetic Hydrochar Grafted-Chitosan for Enhanced Efficient Adsorption of Malachite Green Dye from Aqueous Solutions: Modeling, Adsorption Behavior, and Mechanism Analysis. *Int. J. Biol. Macromol.* **2024**, *254*, 127767. [\[CrossRef\]](#)
38. Delgado-Moreno, L.; Bazhari, S.; Gasco, G.; Méndez, A.; El Azzouzi, M.; Romero, E. New Insights into the Efficient Removal of Emerging Contaminants by Biochars and Hydrochars Derived from Olive Oil Wastes. *Sci. Total Environ.* **2021**, *752*, 141838. [\[CrossRef\]](#) [\[PubMed\]](#)
39. Yudha, S.P.; Tekasakul, S.; Phoungthong, K.; Chuenchom, L. Green Synthesis of Low-Cost and Eco-Friendly Adsorbent for Dye and Pharmaceutical Adsorption: Kinetic, Isotherm, Thermodynamic and Regeneration Studies. *Mater. Res. Express* **2019**, *6*, 125526. [\[CrossRef\]](#)
40. Senthil Kumar, P.; Shanmugapriya, M.; Prasannamedha, G.; Rangasamy, G. Immobilization of Hydrochar in Cellulose Beads for Eradicating Paracetamol from Synthetic and Sewage Water. *Environ. Pollut.* **2024**, *342*, 123035. [\[CrossRef\]](#) [\[PubMed\]](#)
41. Hayoun, B.; Escudero-Curiel, S.; Bourouina, M.; Bourouina-Bacha, S.; Angeles Sanromán, M.; Pazos, M. Preparation and Characterization of High Performance Hydrochar for Efficient Adsorption of Drugs Mixture. *J. Mol. Liq.* **2022**, *353*, 118797. [\[CrossRef\]](#)
42. Petrović, J.; Ercegović, M.; Simić, M.; Koprivica, M.; Dimitrijević, J.; Jovanović, A.; Janković Pantić, J. Hydrothermal Carbonization of Waste Biomass: A Review of Hydrochar Preparation and Environmental Application. *Processes* **2024**, *12*, 207. [\[CrossRef\]](#)
43. Ismail, H.Y.; Shirazian, S.; Skoretska, I.; Mynko, O.; Ghanim, B.; Leahy, J.J.; Walker, G.M.; Kwapiński, W. ANN-Kriging Hybrid Model for Predicting Carbon and Inorganic Phosphorus Recovery in Hydrothermal Carbonization. *Waste Manag.* **2019**, *85*, 242–252. [\[CrossRef\]](#)
44. Cheng, H.; Ji, R.; Bian, Y.; Jiang, X.; Song, Y. From Macroalgae to Porous Graphitized Nitrogen-Doped Biochars—Using Aquatic Biota to Treat Polycyclic Aromatic Hydrocarbons-Contaminated Water. *Bioresour. Technol.* **2020**, *303*, 122947. [\[CrossRef\]](#)
45. Yu, C.; Tang, J.; Su, H.; Huang, J.; Liu, F.; Wang, L.; Sun, H. Development of a Novel Biochar/Iron Oxide Composite from Green Algae for Bisphenol-A Removal: Adsorption and Fenton-like Reaction. *Environ. Technol. Innov.* **2022**, *28*, 102647. [\[CrossRef\]](#)
46. Wang, Y.; Ma, C.; Kong, D.; Lian, L.; Liu, Y. Review on Application of Algae-Based Biochars in Environmental Remediation: Progress, Challenge and Perspectives. *J. Environ. Chem. Eng.* **2023**, *11*, 111263. [\[CrossRef\]](#)
47. Wu, Y.; Cheng, H.; Pan, D.; Zhang, L.; Li, W.; Song, Y.; Bian, Y.; Jiang, X.; Han, J. Potassium Hydroxide-Modified Algae-Based Biochar for the Removal of Sulfamethoxazole: Sorption Performance and Mechanisms. *J. Environ. Manag.* **2021**, *293*, 112912. [\[CrossRef\]](#)
48. Amin, M.; Chetpattananondh, P.; Khan, M.N. Ultrasound Assisted Adsorption of Reactive Dye-145 by Biochars from Marine *Chlorella* Sp. Extracted Solid Waste Pyrolyzed at Various Temperatures. *J. Environ. Chem. Eng.* **2020**, *8*, 104403. [\[CrossRef\]](#)
49. Saber, M.; Takahashi, F.; Yoshikawa, K. Characterization and Application of Microalgae Hydrochar as a Low-Cost Adsorbent for Cu(II) Ion Removal from Aqueous Solutions. *Environ. Sci. Pollut. Res.* **2018**, *25*, 32721–32734. [\[CrossRef\]](#)
50. Leng, L.J.; Yuan, X.Z.; Huang, H.J.; Wang, H.; Wu, Z.B.; Fu, L.H.; Peng, X.; Chen, X.H.; Zeng, G.M. Characterization and Application of Bio-Chars from Liquefaction of Microalgae, Lignocellulosic Biomass and Sewage Sludge. *Fuel Process. Technol.* **2015**, *129*, 8–14. [\[CrossRef\]](#)
51. Parsa, M.; Nourani, M.; Baghdadi, M.; Hosseinzadeh, M.; Pejman, M. Biochars Derived from Marine Macroalgae as a Mesoporous By-Product of Hydrothermal Liquefaction Process: Characterization and Application in Wastewater Treatment. *J. Water Process Eng.* **2019**, *32*, 100942. [\[CrossRef\]](#)
52. Gurav, R.; Bhatia, S.K.; Choi, T.R.; Choi, Y.K.; Kim, H.J.; Song, H.S.; Lee, S.M.; Lee Park, S.; Lee, H.S.; Koh, J.; et al. Application of Macroalgal Biomass Derived Biochar and Bioelectrochemical System with *Shewanella* for the Adsorptive Removal and Biodegradation of Toxic Azo Dye. *Chemosphere* **2021**, *264*, 128539. [\[CrossRef\]](#) [\[PubMed\]](#)
53. Wang, J.; Xie, Q.; Li, A.; Liu, X.; Yu, F.; Ji, J. Biosorption of Hexavalent Chromium from Aqueous Solution by Polyethyleneimine-Modified Ultrasonic-Assisted Acid Hydrochar from *Sargassum Horneri*. *Water Sci. Technol.* **2020**, *81*, 1114–1129. [\[CrossRef\]](#)

54. Kim, H.; Lee, S.Y.; Choi, J.W.; Jung, K.W. Synergistic Effect in Simultaneous Removal of Cationic and Anionic Heavy Metals by Nitrogen Heteroatom Doped Hydrochar from Aqueous Solutions. *Chemosphere* **2023**, *323*, 138269. [\[CrossRef\]](#)
55. Chen, Z.-L.; Xu, H.; Bai, L.Q.; Feng, Y.L.; Li, B. Protonated-Amino-Functionalized Bamboo Hydrochar for Efficient Removal of Hexavalent Chromium and Methyl Orange. *Prog. Nat. Sci. Mater. Int.* **2023**, *33*, 501–507. [\[CrossRef\]](#)
56. Kojić, M.; Mihajlović, M.; Marinović-Cincović, M.; Petrović, J.; Katnić, Đ.; Krstić, A.; Butulija, S.; Onjia, A. Calcium-Pyro-Hydrochar Derived from the Spent Mushroom Substrate as a Functional Sorbent of Pb²⁺ and Cd²⁺ from Aqueous Solutions. *Waste Manag. Res.* **2022**, *40*, 1629–1636. [\[CrossRef\]](#)
57. Ranguin, R.; Delannoy, M.; Yacou, C.; Jean-Marius, C.; Feidt, C.; Rychen, G.; Gaspard, S. Biochar and Activated Carbons Preparation from Invasive *Algae Sargassum* Spp. For Chlordecone Availability Reduction in Contaminated Soils. *J. Environ. Chem. Eng.* **2021**, *9*, 105280. [\[CrossRef\]](#)
58. Jazie, A.A.; Hydary, J.; Abed, S.A.; Al-Dawody, M.F. Hydrothermal Liquefaction of *Fucus Vesiculosus* Algae Catalyzed by H β Zeolite Catalyst for Biocrude Oil Production. *Algal Res.* **2022**, *61*, 102596. [\[CrossRef\]](#)
59. Cavali, M.; Libardi Junior, N.; de Sena, J.D.; Woiciechowski, A.L.; Soccol, C.R.; Belli Filho, P.; Bayard, R.; Benbelkacem, H.; de Castilhos Junior, A.B. A Review on Hydrothermal Carbonization of Potential Biomass Wastes, Characterization and Environmental Applications of Hydrochar, and Biorefinery Perspectives of the Process. *Sci. Total Environ.* **2023**, *857*, 159627. [\[CrossRef\]](#)
60. Martínez-Meraz, C.; González-Fernández, L.A.; Medellín-Castillo, N.A.; López-Cruz, C.M.; Reyes-Hernández, J.; Castillo-Ramos, V.; Sánchez-Polo, M. *Sargassum* Biomass-Derived Biochars for Ibuprofen Removal from Water: Adsorption and Kinetics. *MRS Adv.* **2023**, *8*, 1377–1384. [\[CrossRef\]](#)
61. Hung, C.M.; Chen, C.W.; Huang, C.P.; Dong, C. Di Activation of Peroxymonosulfate by Nitrogen-Doped Carbocatalysts Derived from Brown Algal (*Sargassum Duplicatum*) for the Degradation of Polycyclic Aromatic Hydrocarbons in Marine Sediments. *J. Environ. Chem. Eng.* **2021**, *9*, 106420. [\[CrossRef\]](#)
62. Hung, C.M.; Chen, C.W.; Huang, C.P.; Cheng, J.W.; Dong, C. Di Algae-Derived Metal-Free Boron-Doped Biochar as an Efficient Bioremediation Pretreatment for Persistent Organic Pollutants in Marine Sediments. *J. Clean. Prod.* **2022**, *336*, 130448. [\[CrossRef\]](#)
63. Atugoda, T.; Gunawardane, C.; Ahmad, M.; Vithanage, M. Mechanistic Interaction of Ciprofloxacin on Zeolite Modified Seaweed (*Sargassum Crassifolium*) Derived Biochar: Kinetics, Isotherm and Thermodynamics. *Chemosphere* **2021**, *281*, 130676. [\[CrossRef\]](#)
64. Nguyen, T.B.; Truong, Q.M.; Chen, C.W.; Doong, R.A.; Chen, W.H.; Dong, C. Di Mesoporous and Adsorption Behavior of Algal Biochar Prepared via Sequential Hydrothermal Carbonization and ZnCl₂ Activation. *Bioresour. Technol.* **2022**, *346*, 126351. [\[CrossRef\]](#)
65. Chabi, N.; Baghdadi, M.; Sani, A.H.; Golzary, A.; Hosseinzadeh, M. Removal of Tetracycline with Aluminum Boride Carbide and Boehmite Particles Decorated Biochar Derived from Algae. *Bioresour. Technol.* **2020**, *316*, 123950. [\[CrossRef\]](#) [\[PubMed\]](#)
66. Chen, Y.D.; Lin, Y.C.; Ho, S.H.; Zhou, Y.; Ren, N. qi Highly Efficient Adsorption of Dyes by Biochar Derived from Pigments-Extracted Macroalgae Pyrolyzed at Different Temperature. *Bioresour. Technol.* **2018**, *259*, 104–110. [\[CrossRef\]](#)
67. Ighalo, J.O.; Ohoro, C.R.; Ojukwu, V.E.; Oniye, M.; Shaikh, W.A.; Biswas, J.K.; Seth, C.S.; Mohan, G.B.M.; Chandran, S.A.; Rangabhashiyam, S. Biochar for Ameliorating Soil Fertility and Microbial Diversity: From Production to Action of the Black Gold. *iScience* **2025**, *28*, 11152. [\[CrossRef\]](#) [\[PubMed\]](#)
68. Liu, P.; Rao, D.; Zou, L.; Teng, Y.; Yu, H. Capacity and Potential Mechanisms of Cd(II) Adsorption from Aqueous Solution by Blue Algae-Derived Biochars. *Sci. Total Environ.* **2021**, *767*, 145447. [\[CrossRef\]](#) [\[PubMed\]](#)
69. Moirana, R.L.; Mkunda, J.; Machunda, R.; Paradelo, M.; Mtei, K. Hydroxyapatite-Activated Seaweed Biochar for Enhanced Remediation of Fluoride Contaminated Soil at Various PH Ranges. *Environ. Adv.* **2023**, *11*, 100329. [\[CrossRef\]](#)
70. Yu, H.; Zou, W.; Chen, J.; Chen, H.; Yu, Z.; Huang, J.; Tang, H.; Wei, X.; Gao, B. Biochar Amendment Improves Crop Production in Problem Soils: A Review. *J. Environ. Manag.* **2019**, *232*, 8–21. [\[CrossRef\]](#)
71. Pandey, D.; Singh, S.V.; Savio, N.; Bhutto, J.K.; Srivastava, R.K.; Yadav, K.K.; Sharma, R.; Nandipamu, T.M.K.; Sarkar, B. Biochar Application in Constructed Wetlands for Wastewater Treatment: A Critical Review. *J. Water Process Eng.* **2025**, *69*, 106713. [\[CrossRef\]](#)
72. Waheed, A.; Xu, H.; Qiao, X.; Aili, A.; Yiremaikeybayi, Y.; Haitao, D.; Muhammad, M. Biochar in Sustainable Agriculture and Climate Mitigation: Mechanisms, Challenges, and Applications in the Circular Bioeconomy. *Biomass Bioenergy* **2025**, *193*, 107531. [\[CrossRef\]](#)
73. Liu, Y.N.; He, L.Y. Effects of Alkali-Activated Algae Biochar on Soil Improvement after Phosphorus Absorption: Efficiency and Mechanism. *Sustainability* **2021**, *13*, 11973. [\[CrossRef\]](#)
74. Yang, X.; Jiang, D.; Cheng, X.; Yuan, C.; Wang, S.; He, Z.; Esakkimuthu, S. Adsorption Properties of Seaweed-Based Biochar with the Greenhouse Gases (CO₂, CH₄, N₂O) through Density Functional Theory (DFT). *Biomass Bioenergy* **2022**, *163*, 106519. [\[CrossRef\]](#)
75. Shi, S.; Liu, Y. Nitrogen-Doped Activated Carbons Derived from Microalgae Pyrolysis by-Products by Microwave/KOH Activation for CO₂ Adsorption. *Fuel* **2021**, *306*, 121762. [\[CrossRef\]](#)

76. Chen, H.; Han, X.; Liu, Y. Gaseous Hydrogen Sulfide Removal Using Macroalgae Biochars Modified Synergistically by $\text{H}_2\text{SO}_4/\text{H}_2\text{O}_2$. *Chem. Eng. Technol.* **2021**, *44*, 698–709. [\[CrossRef\]](#)
77. Liu, Z.; Yang, W.; Xu, W.; Liu, Y. Removal of Elemental Mercury by Bio-Chars Derived from Seaweed Impregnated with Potassium Iodine. *Chem. Eng. J.* **2018**, *339*, 468–478. [\[CrossRef\]](#)
78. Balahmar, N.; Al-Jumaily, A.S.; Mokaya, R. Biomass to Porous Carbon in One Step: Directly Activated Biomass for High Performance CO_2 Storage. *J. Mater. Chem. A* **2017**, *5*, 12330–12339. [\[CrossRef\]](#)
79. Wang, Y.; Yang, Q.; Chen, J.; Yang, J.; Zhang, Y.; Chen, Y.; Li, X.; Du, W.; Liang, A.; Ho, S.H.; et al. Adsorption Behavior of Cr(VI) by Magnetically Modified Enteromorpha Prolifera Based Biochar and the Toxicity Analysis. *J. Hazard. Mater.* **2020**, *395*, 122658. [\[CrossRef\]](#)
80. Liu, Z.; Adewuyi, Y.G.; Shi, S.; Chen, H.; Li, Y.; Liu, D.; Liu, Y. Removal of Gaseous Hg_0 Using Novel Seaweed Biomass-Based Activated Carbon. *Chem. Eng. J.* **2019**, *366*, 41–49. [\[CrossRef\]](#)
81. Wang, H.; Wang, H.; Zhao, H.; Yan, Q. Adsorption and Fenton-like Removal of Chelated Nickel from Zn-Ni Alloy Electroplating Wastewater Using Activated Biochar Composite Derived from Taihu Blue Algae. *Chem. Eng. J.* **2020**, *379*, 122372. [\[CrossRef\]](#)
82. Araya, M.; Rivas, J.; Sepúlveda, G.; Espinoza-González, C.; Lira, S.; Meynard, A.; Blanco, E.; Escalona, N.; Ginocchio, R.; Garrido-Ramírez, E.; et al. Effect of Pyrolysis Temperature on Copper Aqueous Removal Capability of Biochar Derived from the Kelp *Macrocystis Pyrifera*. *Appl. Sci.* **2021**, *11*, 9223. [\[CrossRef\]](#)
83. Khanzada, A.K.; Rizwan, M.; Al-Hazmi, H.E.; Majtacz, J.; Kurniawan, T.A.; Makinia, J. Removal of Arsenic from Wastewater Using Hydrochar Prepared from Red Macroalgae: Investigating Its Adsorption Efficiency and Mechanism. *Water* **2023**, *15*, 3866. [\[CrossRef\]](#)
84. Wang, C.; Li, X.; Wu, W.; Chen, G.; Tao, J. Removal of Cadmium in Water by Potassium Hydroxide Activated Biochar Produced from Enteromorpha Prolifera. *J. Water Process Eng.* **2021**, *42*, 102201. [\[CrossRef\]](#)
85. Li, X.; Wang, C.; Tian, J.; Liu, J.; Chen, G. Comparison of Adsorption Properties for Cadmium Removal from Aqueous Solution by Enteromorpha Prolifera Biochar Modified with Different Chemical Reagents. *Environ. Res.* **2020**, *186*, 109502. [\[CrossRef\]](#)
86. Kumar, P.; Patel, A.K.; Singhania, R.R.; Chen, C.W.; Saratale, R.G.; Dong, C. Di Enhanced Copper (II) Bioremediation from Wastewater Using Nano Magnetite (Fe_3O_4) Modified Biochar of *Ascophyllum Nodosum*. *Bioresour. Technol.* **2023**, *388*, 129654. [\[CrossRef\]](#)
87. Tan, L.; Nie, Y.; Chang, H.; Zhu, L.; Guo, K.; Ran, X.; Zhong, N.; Zhong, D.; Xu, Y.; Ho, S.H. Adsorption Performance of Ni(II) by KOH-Modified Biochar Derived from Different Microalgae Species. *Bioresour. Technol.* **2024**, *394*, 130287. [\[CrossRef\]](#)
88. Kumar, M.; Singh, A.K.; Sikandar, M. Biosorption of Hg (II) from Aqueous Solution Using Algal Biomass: Kinetics and Isotherm Studies. *Heliyon* **2020**, *6*, 1465–1469. [\[CrossRef\]](#) [\[PubMed\]](#)
89. Moon, S.; Lee, Y.J.; Choi, M.Y.; Lee, C.G.; Park, S.J. Adsorption of Heavy Metals and Bisphenol A from Wastewater Using Spirulina Sp.-Based Biochar as an Effective Adsorbent: A Preliminary Study. *J. Appl. Phycol.* **2023**, *35*, 2257–2269. [\[CrossRef\]](#)
90. Yang, Z.; Hou, J.; Wu, J.; Miao, L. The Effect of Carbonization Temperature on the Capacity and Mechanisms of Pb(II) Adsorption by Microalgae Residue-Derived Biochar. *Ecotoxicol. Environ. Saf.* **2021**, *225*, 112750. [\[CrossRef\]](#) [\[PubMed\]](#)
91. Yaashikaa, P.R.; Keerthana Devi, M.; Senthil Kumar, P.; Pandian, E. A Review on Biodiesel Paroduction by Algal Biomass: Outlook on Lifecycle Assessment and Techno-Economic Analysis. *Fuel* **2022**, *324*, 124774. [\[CrossRef\]](#)
92. Kumar, S.; Tirlangi, S.; Kumar, A.; Imran, M.; Pillai HP, J.S.; Kumar Koshariya, A.; Sathish, T.; Ubaidullah, M.; Ayub, R.; Minnam Reddy, V.R.; et al. A Review on the Contribution of Nanotechnology for Biofuel Production from Algal Biomass: A Bridge to the Reduction of Carbon Footprint. *Sustain. Energy Technol. Assess.* **2023**, *60*, 103498. [\[CrossRef\]](#)
93. Li, G.; Yao, J. A Review of Algae-Based Carbon Capture, Utilization, and Storage (Algae-Based CCUS). *Gases* **2024**, *4*, 468–503. [\[CrossRef\]](#)
94. Zhang, Q.; Guan, Y.; Zhang, Z.; Dong, S.; Yuan, T.; Ruan, Z.; Chen, M. Sustainable Microalgae Cultivation: A Comprehensive Review of Open and Enclosed Systems for Biofuel and High Value Compound Production. In *E3S Web of Conferences*; EDP Sciences: Les Ulis, France, 2024; Volume 577, p. 1008.
95. Ijaola, A.O.; Akamo, D.O.; George, T.T.; Sengul, A.; Adediji, M.Y.; Asmatulu, E. Algae as a Potential Source of Protein: A Review on Cultivation, Harvesting, Extraction, and Applications. *Algal Res.* **2024**, *77*, 103329. [\[CrossRef\]](#)
96. Sun, J.; Norouzi, O.; Mašek, O. A State-of-the-Art Review on Algae Pyrolysis for Bioenergy and Biochar Production. *Bioresour. Technol.* **2022**, *346*, 126258. [\[CrossRef\]](#)
97. Kannah, R.Y.; Kavitha, S.; Banu, J.R.; Sivashanmugam, P.; Gunasekaran, M.; Kumar, G. A Mini Review of Biochemical Conversion of Algal Biorefinery. *Energy Fuels* **2021**, *35*, 16995–17007. [\[CrossRef\]](#)
98. Poornima, S.; Manikandan, S.; Prakash, R.; Deena, S.R.; Subbaiya, R.; Karmegam, N.; Kim, W.; Govarthanan, M. Biofuel and Biochemical Production through Biomass Transformation Using Advanced Thermochemical and Biochemical Processes—A Review. *Fuel* **2024**, *372*, 132204. [\[CrossRef\]](#)

99. Farouk, S.M.; Tayeb, A.M.; Abdel-Hamid, S.M.S.; Osman, R.M. Recent Advances in Transesterification for Sustainable Biodiesel Production, Challenges, and Prospects: A Comprehensive Review. *Environ. Sci. Pollut. Res.* **2024**, *31*, 12722–12747. [[CrossRef](#)] [[PubMed](#)]
100. Bassoli, S.C.; da Fonseca, Y.A.; Wandurraga, H.J.L.; Baeta, B.E.L.; de Souza Amaral, M. Research Progress, Trends, and Future Prospects on Hydrothermal Liquefaction of Algae for Biocrude Production: A Bibliometric Analysis. *Biomass Convers. Biorefinery* **2023**, *14*, 28781–28796. [[CrossRef](#)] [[PubMed](#)]
101. El-Araby, R. Biofuel Production: Exploring Renewable Energy Solutions for a Greener Future. *Biotechnol. Biofuels Bioprod.* **2024**, *17*, 129. [[CrossRef](#)]
102. Quiroz, D.; McGowen, J.; Quinn, J.C. Implications of Pond Reliability on the Techno-Economic and Life Cycle Environmental Impacts of Algal Biofuels. *J. Clean. Prod.* **2024**, *469*, 143178. [[CrossRef](#)]
103. Zarei, Z.; Malekshahi, P.; Morowvat, M.H.; Trzcinski, A.P. Hydrodynamic Performance of Algal Bioreactors. *Algal Bioreact. Sci. Eng. Technol. Upstream Process.* **2024**, *1*, 121–133. [[CrossRef](#)]
104. Pandey, S.; Narayanan, I.; Selvaraj, R.; Varadavenkatesan, T.; Vinayagam, R. Biodiesel Production from Microalgae: A Comprehensive Review on Influential Factors, Transesterification Processes, and Challenges. *Fuel* **2024**, *367*, 131547. [[CrossRef](#)]
105. Yang, B.; Liu, J.; Ma, X.; Guo, B.; Liu, B.; Wu, T.; Jiang, Y.; Chen, F. Genetic Engineering of the Calvin Cycle toward Enhanced Photosynthetic CO₂ Fixation in Microalgae. *Biotechnol. Biofuels* **2017**, *10*, 229. [[CrossRef](#)] [[PubMed](#)]
106. Hlavova, M.; Turoczy, Z.; Bisova, K. Improving Microalgae for Biotechnology—From Genetics to Synthetic Biology. *Biotechnol. Adv.* **2015**, *33*, 1194–1203. [[CrossRef](#)]
107. Brar, A.; Kumar, M.; Soni, T.; Vivekanand, V.; Pareek, N. Insights into the Genetic and Metabolic Engineering Approaches to Enhance the Competence of Microalgae as Biofuel Resource: A Review. *Bioresour. Technol.* **2021**, *339*, 125597. [[CrossRef](#)]
108. Patel, V.K.; Das, A.; Kumari, R.; Kajla, S. Recent Progress and Challenges in CRISPR-Cas9 Engineered Algae and Cyanobacteria. *Algal Res.* **2023**, *71*, 103068. [[CrossRef](#)]
109. Joshi, S.; Mishra, S.D. Recent Advances in Biofuel Production through Metabolic Engineering. *Bioresour. Technol.* **2022**, *352*, 127037. [[CrossRef](#)]
110. Valizadeh Derakhshan, M.; Nasernejad, B.; Abbaspour-Aghdam, F.; Hamidi, M. Oil Extraction from Algae: A Comparative Approach. *Biotechnol. Appl. Biochem.* **2015**, *62*, 375–382. [[CrossRef](#)] [[PubMed](#)]
111. Rathinavel, L.; Ravikumar, Y.; Jothinathan, D.; Pandey, A.; Mahata, C. Extraction and Enrichment of Fatty Acids from Marine Microalgae. In *Marine Molecules from Algae and Cyanobacteria: Extraction, Purification, Toxicology and Applications*; Elsevier: Amsterdam, The Netherlands, 2024; pp. 41–57. ISBN 9780443216749.
112. Ezhumalai, G.; Arun, M.; Manavalan, A.; Rajkumar, R.; Heese, K. A Holistic Approach to Circular Bioeconomy Through the Sustainable Utilization of Microalgal Biomass for Biofuel and Other Value-Added Products. *Microb. Ecol.* **2024**, *87*, 61. [[CrossRef](#)]
113. Gentili, F.G.; Fick, J. Algal Cultivation in Urban Wastewater: An Efficient Way to Reduce Pharmaceutical Pollutants. *J. Appl. Phycol.* **2017**, *29*, 255–262. [[CrossRef](#)] [[PubMed](#)]
114. Bakonyi, P.; Peter, J.; Koter, S.; Mateos, R.; Kumar, G.; Koók, L.; Rózsenszki, T.; Pientka, Z.; Kujawski, W.; Kim, S.H.; et al. Possibilities for the Biologically-Assisted Utilization of CO₂-Rich Gaseous Waste Streams Generated during Membrane Technological Separation of Biohydrogen. *J. CO₂ Util.* **2020**, *36*, 231–243. [[CrossRef](#)]
115. Comley, J.G.; Scott, J.A.; Laamanen, C.A. Utilizing CO₂ in Industrial Off-Gas for Microalgae Cultivation: Considerations and Solutions. *Crit. Rev. Biotechnol.* **2024**, *44*, 910–923. [[CrossRef](#)] [[PubMed](#)]
116. Vieira Costa, J.A.; Cruz, C.G.; Centeno da Rosa, A.P. Insights into the Technology Utilized to Cultivate Microalgae in Dairy Effluents. *Biocatal. Agric. Biotechnol.* **2021**, *35*, 102106. [[CrossRef](#)]
117. Zewdie, D.T.; Ali, A.Y. Cultivation of Microalgae for Biofuel Production: Coupling with Sugarcane-Processing Factories. *Energy. Sustain. Soc.* **2020**, *10*, 27. [[CrossRef](#)]
118. Thomas, P.K.; Dunn, G.P.; Coats, E.R.; Newby, D.T.; Feris, K.P. Algal Diversity and Traits Predict Biomass Yield and Grazing Resistance in Wastewater Cultivation. *J. Appl. Phycol.* **2019**, *31*, 2323–2334. [[CrossRef](#)]
119. Sathinathan, P.; Parab, H.M.; Yusoff, R.; Ibrahim, S.; Vello, V.; Ngoh, G.C. Photobioreactor Design and Parameters Essential for Algal Cultivation Using Industrial Wastewater: A Review. *Renew. Sustain. Energy Rev.* **2023**, *173*, 113096. [[CrossRef](#)]
120. Hekmatmehr, H.; Esmaeili, A.; Pourmahdi, M.; Atashrouz, S.; Abedi, A.; Ali Abuswer, M.; Nedeljkovic, D.; Latifi, M.; Farag, S.; Mohaddespour, A. Carbon Capture Technologies: A Review on Technology Readiness Level. *Fuel* **2024**, *363*, 130898. [[CrossRef](#)]
121. Yang, J.; Zhao, T.; Cui, X.; Peng, M.; Wang, X.; Mao, H.; Cui, M. New Insights into the Carbon Neutrality of Microalgae from Culture to Utilization: A Critical Review on the Algae-Based Solid Biofuels. *Biomass Bioenergy* **2022**, *166*, 106599. [[CrossRef](#)]
122. Chia, S.R.; Nomanbhay, S.B.H.M.; Chew, K.W.; Munawaroh, H.S.H.; Shamsuddin, A.H.; Show, P.L. Algae as Potential Feedstock for Various Bioenergy Production. *Chemosphere* **2022**, *287*, 131944. [[CrossRef](#)] [[PubMed](#)]
123. Jiang, Y.; Zhou, G.; Zhang, H.; Xu, J.; Ge, H.; Shen, L.; Song, T. Coupling Effects of Heating Pelletizing and Torrefaction on Black Pellets Production from Microalga *Nannochloropsis Oceanica* Residues. *Fuel* **2023**, *353*, 129007. [[CrossRef](#)]

124. Kosowska-Golachowska, M.; Musiał, T.; Urbaniak, D.; Otwinowski, H. Analysis of Microalgae Pellets Combustion in a Circulating Fluidized-Bed. In *E3S Web of Conferences*; EDP Sciences: Les Ulis, France, 2017; Volume 14, p. 2035.
125. Miranda, M.T.; Sepúlveda, F.J.; Arranz, J.I.; Montero, I.; Rojas, C.V. Physical-Energy Characterization of Microalgae *Scenedesmus* and Experimental Pellets. *Fuel* **2018**, *226*, 121–126. [[CrossRef](#)]
126. Kastanaki, E.; Vamvuka, D. A Comparative Reactivity and Kinetic Study on the Combustion of Coal-Biomass Char Blends. *Fuel* **2006**, *85*, 1186–1193. [[CrossRef](#)]
127. Gong, Z.; Wang, Z.; Wang, Z.; Du, A.; Fang, P.; Sun, Z.; Li, X. Study on Pyrolysis of Oil Sludge with Microalgae Residue Additive. *Can. J. Chem. Eng.* **2018**, *96*, 1919–1925. [[CrossRef](#)]
128. Wu, X.; Liu, J.; Wei, Z.; Chen, Z.; Evrendilek, F.; Huang, W. Oxy-Fuel Co-Combustion Dynamics of Phytoremediation Biomass and Textile Dyeing Sludge: Gas-to-Ash Pollution Abatement. *Sci. Total Environ.* **2022**, *825*, 153656. [[CrossRef](#)] [[PubMed](#)]
129. Chen, C.; Chen, F.; Cheng, Z.; Chan, Q.N.; Kook, S.; Yeoh, G.H. Emissions Characteristics of NO_x and SO₂ in the Combustion of Microalgae Biomass Using a Tube Furnace. *J. Energy Inst.* **2017**, *90*, 806–812. [[CrossRef](#)]

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.